Pecuniary Externality in the (Competitive) OTC Market

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ABSTRACT

A competitive over-the-counter market gives an otherwise illiquid asset *indirect liquidity*: a buyer can sell it for cash exactly when she wants to spend, without the asset ever serving as a medium of exchange. We show that this liquidity service creates a pecuniary externality once the goods market is one in which sellers post prices and identical buyers sample different quotes. Buyers who convert bring more purchasing power to the goods market, firms with market power respond by posting higher prices, and *the buyers who never enter the asset market are the ones who pay them*. At low rates of inflation the externality outweighs the liquidity it comes with, so the asset market lowers welfare, and it lowers welfare by more the more plentiful the asset. Endogenizing the asset supply confirms the pattern: competitive issuers put more of the asset into circulation than a monopolist would, so the welfare loss is larger under competitive issuance.

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1 Introduction

This paper asks what a competitive over-the-counter asset market does to the distribution of prices at which buyers trade, and what that does to welfare. The market in question performs a service the literature calls indirect liquidity: it lets a buyer sell an asset for cash at the moment she wants to spend, so that the asset is liquid even though it never changes hands in a goods trade. We show that this service, valuable as it is to the buyer who uses it, creates a pecuniary externality on the buyer who does not. Sellers with market power observe that buyers can raise cash and respond by posting higher prices, and the buyer who ends up paying out of her own money is quoted those prices without ever having entered the asset market. At low rates of inflation the loss she suffers outweighs the gain to those who convert, so the asset market lowers welfare rather than raising it, and it does so by more the more plentiful the asset is.

The model combines two blocks, each standard on its own. The first is the price-posting economy of Head et al. (2012), which brings Burdett and Judd (1983) into a monetary environment: sellers post prices, buyers sample a random number of quotes and buy at the lowest one they see, and the equilibrium is a non-degenerate distribution of prices across sellers who are identical. The second is the over-the-counter asset market of Geromichalos and Herrenbrueck (2016), in the competitive version of their appendix: households leave the centralized market holding money and an asset, an over-the-counter market convenes before goods trade, and buyers who want more cash sell the asset to buyers who do not. The asset pays a dividend in the next centralized market and cannot be handed over in a goods trade, so whatever liquidity it has is indirect. Both blocks sit inside the quasi-linear framework of Lagos and Wright (2005), so the distribution of wealth resets each period and the equilibrium object is a pair: the real balances buyers carry and the price at which the asset changes hands in the secondary market.

What the combination adds is that the two sides of the asset market are made by the goods market rather than given to it. In the environments this literature usually works with, a buyer learns whether she wants to consume before she visits the secondary market, and that shock is exogenous; here every buyer wants to consume, and what separates them is the price they happen to draw. A buyer who samples a cheap quote finds her cash binding and wants to sell the asset; a buyer who samples an expensive one does not, and is content to lend her idle cash instead. Search noise therefore decides who borrows and who lends, and the measures on the two sides are equilibrium objects that move with monetary policy and with the supply of the asset. It also means that only some buyers ever want to convert, which is the feature that makes the externality possible: there is a group that pays the prices the asset market produces without taking any part in producing them.

The externality works through a wedge between what a unit of cash is worth to a buyer who can sell assets and to one who cannot. When the asset is scarce, lenders bring more cash than borrowers can absorb, the secondary market clears at the dividend, and the asset is a perfect substitute for money: it displaces real balances one for one and the distribution of posted prices is exactly the one that would prevail with no asset market at all. That case is worth stating on its own, because it shows that an asset market which is open, active, and carrying its largest possible liquidity premium can leave the goods market untouched. Once the asset is plentiful enough that lenders cannot absorb it at the dividend, its price falls and selling becomes costly at the margin; that cost is the wedge, and everything else in the paper follows from it. Buyers who draw cheap quotes can now spend more than their own money would allow, buyers who draw expensive quotes have less of their own money to spend, and sellers, seeing both, shift the distribution of posted prices upward.

Whether that shift lowers welfare is settled by the rate of inflation, and at low rates it does. Two forces work against each other: the asset supplies liquidity, which is worth something to a buyer who is short of it, and it raises the prices at which she spends, which is worth something to the seller instead. When inflation is low, money is cheap to hold, buyers carry a great deal of it, and the quantities they buy are already close to the ones a planner would choose, so the extra liquidity is worth very little at the margin; the price effect, by contrast, is governed by the wedge and does not care how close to efficiency the buyer already was. The externality therefore wins exactly where the private gain from the asset market is smallest, and the surplus generated in the goods market falls. At the Friedman rule the effect disappears altogether, not because the asset market closes but because the interval in which its price can lie collapses onto the dividend, and at high rates of inflation the sign reverses, since buyers are by then constrained badly enough that the liquidity is worth more than the higher prices cost them.

The size of the effect grows with the supply of the asset, and it does so over a bounded range. Below a first cutoff nothing happens at all: the asset trades at its dividend, the two economies coincide, and the deviation from the benchmark is exactly zero at every rate of inflation. Above that cutoff the secondary-market price falls, the wedge opens, and each additional unit of the asset pushes that price down further and the distribution of goods prices up further, so the welfare loss deepens as the asset becomes more plentiful. Above a second cutoff no buyer is ever capped by her whole portfolio, the asset supply drops out of the equilibrium conditions, and everything freezes: a larger supply changes nothing and the loss stops growing. Inflation moves both cutoffs down, so a fixed quantity of the asset can pass from the first regime into the second as inflation rises, which is to say that monetary policy decides whether the externality is present and the asset supply decides how large it is.

Letting the supply be chosen sharpens the result rather than softening it. An issuer's profit per unit is the liquidity premium on the asset, so a monopolist's problem contains the very price effect the externality consists of: he knows that another unit depresses the price at which he sells every unit he has already issued, and he stops before the asset market begins to operate. At his choice the externality is exactly zero and the distribution of prices is the one that would prevail with no asset market at all, even though the asset exists, is held, and is traded. Competitive issuers, for whom the asset price is a number rather than something their own decisions move, enter until the premium is exhausted, and in doing so they put more of the asset into circulation than a monopolist would. The externality is therefore larger under competitive issuance than under monopoly, and the two market structures turn out to be the two endpoints of the interval that contains the choice a constrained planner would make.

Three contributions follow. The first is to show that the liquidity service of a competitive over-the-counter market is not neutral for the goods market but is itself a source of pecuniary externality, operating through the price at which the asset is resold rather than through a rate at which credit is extended. The second is to show that the externality is ordered by the abundance of the asset: it is absent while the asset is scarce, deepens as the asset becomes plentiful, and stops once the asset ceases to be scarce, a comparative static that has no counterpart in environments where the goods market clears at a single price. The third is that the same ordering survives when the supply is chosen rather than given, and that it inverts the usual reading of market power in issuance, since monopoly internalizes the price effect precisely because that effect is the monopolist's revenue while free entry ignores it and maximizes the externality. Taken together these say that the liquidity of an asset and the dispersion of goods prices are not separate questions, and that policy towards the supply of liquid assets cannot be settled without knowing the rate of inflation at which it will operate.

Related literature. Four strands meet here. The first is the literature on indirect liquidity, in which an asset that cannot be handed over in a goods trade is nonetheless liquid because it can be sold for cash before that trade takes place. Geromichalos and Herrenbrueck (2016) is the model we build on: money and an illiquid asset are chosen in advance, a consumption opportunity arrives, and a secondary market lets buyers rebalance. Their main text prices that market by search and bargaining, following Duffie et al. (2005); the competitive version in their appendix is the one we adopt. The same indirect channel appears in Benjamin Lester (2012), Li and Li (2013) and He et al. (2015). In all of this work the goods market clears at a single price, so the only thing the secondary market can move is the *level* of liquidity. *Our departure is to place that market in front of a goods market that generates a distribution of prices, and to ask what it does to the distribution.* It widens the support downward and moves the mass up, at the expense of buyers who never enter the asset market. Geromichalos et al. (2023) endogenize the supply of liquid assets in such an environment and supply the issuer's problem we use in Section 4; there too the goods market clears at one price, so the issuer's decision has no counterpart in dispersion, whereas here it selects both the size of the externality and whether there is one at all.

The second strand is price posting with equilibrium dispersion. Our goods market is Head et al. (2012), which brings Burdett and Judd (1983) into a monetary economy and is the source of the equilibrium price distribution we work with. Head and Kumar (2005) and Wang (2016) study inflation and dispersion in related environments and Camera and Corbae (1999) is an early treatment, while Sheremirov (2019) documents the empirical relation between inflation and dispersion that this class of models speaks to. In all of them the only asset is money, so a buyer's liquidity is whatever she chose to carry; here she has a second margin, and the distribution responds to it.

The third strand, and the one this paper speaks to most directly, concerns liquidity, market power, and the externality the two together create. The externality we identify is the one in Kam et al. (2024). There, buyers with access to competitive banking bring more purchasing power to the goods market, firms with market power respond by shifting the price distribution up, and the buyers who do not borrow pay for it. Ours has the same shape, with one difference that runs through everything: their instrument is credit, created on demand at a rate, while ours is an asset in fixed supply carrying a price of its own. *That price, and not the quantity of the instrument, is what the externality tracks*, which is why the effect here switches on at a threshold in the asset supply, deepens as the asset becomes more plentiful, and then stops. It is also why the supply can be made a choice, which is what Section 4 does. The imperfectly competitive credit market of Head et al. (2025) is a related treatment of liquidity and market power.

The fourth strand is the wider literature on pecuniary externalities from liquidity provision. Chiu et al. (2018) show that credit can impose a pecuniary-externality cost on liquidity-constrained buyers even under competitive goods and banking markets, though their result requires an exogenous measure of constrained buyers and a strictly convex production cost; here the measures of buyer types are equilibrium objects and costs are linear. Berentsen et al. (2007) and Berentsen et al. (2014) are the benchmark in which intermediation reallocates idle money and is unambiguously good, a channel present in our model as well and one that dominates when the contact rate is low. Dong and Huangfu (2021) obtain a welfare-reducing role for credit through the cost of using it rather than through prices. Geromichalos and Herrenbrueck (2016) note an externality of the opposite sign in their own setting, since a buyer's money holdings insure not only her own consumption but also the trades of those who buy assets from her.

Plan. Section 2 sets out the model and characterizes equilibrium. Section 3 compares the economy with and without the asset market, first across asset supplies and then across inflation rates. Section 4 endogenizes the asset supply. Section 5 concludes.

2 The Model

The model is a Lagos–Wright economy with two additions: a decentralized goods market in which sellers post prices, and an over-the-counter market that convenes between the centralized market and the goods market. Section 2.1 sets out the environment and the order in which markets meet. Section 2.2 takes the household through a period and derives the buyer's demand for goods together with her use of the asset market, which is Lemma 1 and the object everything later is built on. Section 2.3 prices the asset market and shows that both of its sides are made by the goods market rather than given to it. Section 2.4 turns to the firm: it locates the highest price anyone posts, Lemma 2, then the whole distribution of posted prices, Lemma 3, and records how that distribution moves with the buyer's real balances, Lemma 4. Section 2.5 closes the model, states money demand and asset demand as Lemma 5, and shows that the equilibrium falls into one of three regimes according to how much of the asset there is.

2.1 Environment

Timing and notation. Time is discrete and continues forever. Agents discount across periods by a common factor $\beta \in (0, 1)$, but there is no discounting between submarkets within a period. We write $X \equiv X_t$ and $X_+ \equiv X_{t+1}$, and use a hat to denote next period's value of a price, so that $\hat{\varphi} = \varphi_+$.

There is a unit measure of buyers and a unit measure of firms, and buyers are *ex ante* identical. Firms post prices in the decentralized goods market; they are owned by households and rebate their profits in the centralized market, as in Head et al. (2012). Two assets are traded: fiat money, whose supply grows at the rate $\mu > \beta - 1$ through lump-sum transfers, and a real asset in fixed supply $A > 0$ that pays a dividend d in the following centralized market. Buyers are anonymous in the decentralized goods market and private promises cannot be enforced, so unsecured credit is infeasible and money is essential. The real asset cannot be handed over for goods, because sellers cannot verify the authenticity of objects other than money; it acquires an indirect liquidity value only through its convertibility into money.

The order of markets. Within each period, markets convene in the following order.

1. **The centralized market (CM).** Buyers consume the general good, supply labor, collect asset dividends and the monetary transfer, and rebalance their portfolios into money $\hat{m}$ and assets $\hat{a}$ at the prices φ and ψ. The CM then closes.

2. **The decentralized market opens: search and price observation.** Firms post nominal prices. Each buyer contacts k firms with probability α_k and draws k price quotes independently from the posted-price distribution F; following Head et al. (2012) a buyer samples at most two quotes and purchases at the lower of them. At this point a buyer observes her own quotes—and therefore learns whether her money holdings will bind—but no goods have yet changed hands.

3. **The over-the-counter market (OTC) opens.** Assets are exchanged for money at the competitive secondary-market price ψ_C, which agents take as given under rational expectations. A buyer who drew a sufficiently low quote is liquidity constrained and sells assets for cash: she is a *borrower*. A buyer who contacted no firm cannot trade in this period's goods market, so the money she carries is idle; she supplies that cash and acquires assets in order to collect the dividend in the next CM: she is a *lender*. The OTC market then closes.

4. **The decentralized market closes: goods trade.** Buyers holding a quote purchase the decentralized good at the posted price, paying with their own money together with any cash raised in the OTC market. Firms produce on the spot at constant marginal cost. The DM then closes and the next period begins.

The asset market convenes after buyers have observed their quotes because buyers are *ex*

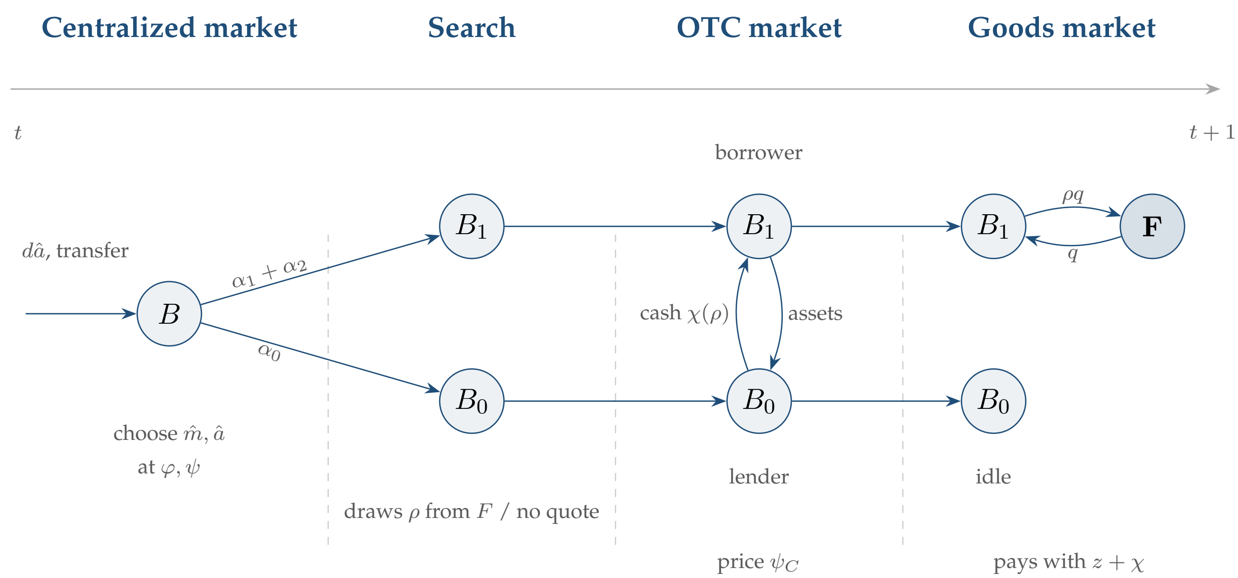


Figure 1: The order of markets within a period.

ante identical: it is search, not an exogenous preference shock, that creates the two sides of the market, and under dispersed prices a buyer does not know the size of her cash shortfall until she samples.[1] One consequence is worth recording. The supply of cash to the asset market is governed by α_0, while the demand for it depends on the realized price and hence on the whole distribution F; both sides are therefore equilibrium objects that respond to monetary policy.

Agents. There are two permanently distinct types of agents, buyers and firms, named for the roles they play in the decentralized market. All agents live forever and discount across periods, but not across sub-periods, at the rate $\beta \in (0,1)$. The measure of buyers is normalized to unity. The measure of firms plays no role beyond the contact probabilities introduced below, since there is no free entry. Buyers consume in both goods markets and supply labor in the centralized market; firms consume only in the centralized market and produce in both. There is no consumption and no production in the over-the-counter round: its only role is to let agents rebalance their portfolios before goods change hands.

Preferences and technology. A buyer's period preferences are given by $\mathcal{U}(q, x, h)$ and a firm's by $\mathcal{V}(x, h, y)$, where x and h are consumption and labor in the centralized market, q is a buyer's

[1]This is where our timing departs from Geromichalos and Herrenbrueck (2016), in whose model an exogenous liquidity shock partitions buyers before the asset market opens and the goods market clears at a single price. It follows Kam et al. (2024), who place a competitive credit market inside the decentralized market in the same way.

consumption in the decentralized market, and y is a firm's production there. Following Lagos and Wright we adopt the functional forms

$$\mathcal{U}(q,x,h) = u(q) + U(x) - h,$$
$$\mathcal{V}(x,h,y) = x - h - c\,y. \tag{2.1}$$

We assume $u' > 0$, $u'' < 0$, and that u satisfies the standard Inada conditions, and likewise for U. We restrict attention to the constant-relative-risk-aversion class,

$$u(q) = \frac{q^{1-\sigma}}{1-\sigma}, \qquad \sigma > 0, \tag{2.2}$$

so that σ governs the price elasticity of a buyer's demand, which is $1/\sigma$. Throughout we impose $\sigma < 1$, and the restriction does two things, one inherited from each half of the model. As in Head et al. (2012), it makes the elasticity $1/\sigma$ exceed unity, so that the price a monopolist would charge an unconstrained buyer, $c/(1-\sigma)$, is finite. As in Head et al. (2025) and Kam et al. (2024), it orders the cutoff prices at which a buyer switches between financing margins; the ordering reverses if $\sigma > 1$. Both papers report that calibrating a model of this class to aggregate money demand delivers σ below unity.

One unit of labor produces one unit of the general good, which is the numeraire, and firms produce the decentralized good on the spot at constant marginal cost c.

Matching in the decentralized market. As in Head et al. (2012), a buyer samples k price quotes with probability α_k, where $\alpha_0 + \alpha_1 + \alpha_2 = 1$ with $\alpha_1 > 0$ and $\alpha_2 > 0$, and purchases at the lower of the quotes she holds. Both α_1 and α_2 must be strictly positive for the posted-price distribution to be non-degenerate: with $\alpha_2 = 0$ every firm faces a captive buyer and posts the monopoly price, while with $\alpha_1 = 0$ price competition drives every firm to marginal cost. A buyer who samples no quote trades no goods this period, and it is her idle money that will supply the asset market.

Money. Money has no intrinsic value, but it is portable, storable, divisible, and recognizable by all agents, so it can serve as a medium of exchange in the decentralized market. Its stock grows at the constant rate $\mu > \beta - 1$ through lump-sum transfers to buyers in the centralized market, and φ denotes its price in units of the general good. In a stationary equilibrium $\varphi/\hat{\varphi} = 1 + \mu$, and the cost of holding money is summarized by the nominal interest rate on an illiquid bond, $i \equiv (1+\mu)/\beta - 1 > 0$. Because carrying money is costly, firms leave the centralized market holding neither money nor assets.

The asset. The asset is a one-period, riskless, real claim: a unit bought in the centralized market at price ψ pays $d > 0$ units of the general good in the following centralized market and is then retired. A quantity A is issued each period; in the benchmark A is a parameter, and Section 4 lets an issuer choose it. Holdings are non-negative, so no agent can sell the asset short. Absent any liquidity role the asset would be priced at its fundamental value,

$$\psi^f = \beta d, \tag{2.3}$$

and the object of interest throughout is the wedge $\psi - \psi^f$. We write ψ_C for the price at which the asset trades against money in the over-the-counter round, also expressed in units of the general good, and reserve ψ for the centralized-market price.

2.2 The household's problem

We work forward through the sub-periods, starting in the centralized market. The buyer's portfolio choice there fixes the pair $(z, \hat{a})$ she carries out; search then splits her, and the two resulting problems — the one solved by a buyer who drew no quote and the one solved by a buyer who drew a quote — are the supply and the demand sides of the asset market. Price posting is taken up in Section 2.4 and the clearing of the two markets in Section 2.5.

Throughout, the buyer is a price taker in every market she enters. She takes the centralized-market prices φ and ψ as given, she takes the secondary-market price ψ_C as given, since that market is competitive, and she takes the quote ρ she has drawn as given, since firms post and do not bargain. What she knows when she chooses her portfolio is the distribution F from which quotes are drawn, not the quote itself; she learns that only after the centralized market has closed. Rational expectations here means that the F she uses is the one firms generate in Section 2.4.

Nominal money is not a convenient unit to work in, and two different dates are involved, so we fix the convention once. A buyer arrives in the centralized market holding m units of money, whose value *today* is φm. What she chooses is how much money to carry *out*, and we measure that at the value it will have when it is spent, so that $z \equiv \hat{\varphi}\hat{m}$ and $\rho \equiv \hat{\varphi}p$ denote real balances and the real price corresponding to a posted nominal price p. The two are linked by $\varphi/\hat{\varphi} = 1 + \mu$ in a stationary equilibrium, so acquiring z costs $(1 + \mu)z$ units of today's general good. Everything the buyer decides is therefore expressed in the goods of the following centralized market; only the wealth she walks in with, $\varphi m + d\,a$, is expressed in today's.

The centralized market

A buyer who enters the centralized market with holdings (m, a) solves

$$W(m, a) = \max_{x,h,z,\hat{a}} \Big\{ U(x) - h + \mathbb{E}\big[\Omega(z, \hat{a})\big] \Big\} \tag{2.4}$$

subject to the budget constraint

$$x + (1 + \mu)z + \psi\hat{a} \;=\; h + \varphi\big(m + T\big) + d\,a, \tag{2.5}$$

where T is the lump-sum transfer and $\mathbb{E}[\Omega]$ is the expected value of entering the decentralized market, described below. The left-hand side is what she spends today — consumption, the money she carries out, and the assets she buys — and the right-hand side is what she has: labor income, the money she walked in with together with the transfer, and the dividend on the assets she walked in with.

At an optimum x and h are separately indeterminate but their difference is not. Writing x^* for the solution to $U'(x^*) = 1$ and substituting $x - h$ from (2.5) into (2.4) gives

$$W(m, a) = \varphi m + d\,a + \Upsilon, \tag{2.6}$$

where the constant Υ collects everything that does not depend on the state,

$$\Upsilon \;\equiv\; U(x^*) - x^* + \varphi T \;+\; \max_{z,\hat{a}} \Big\{ -(1 + \mu)z - \psi\hat{a} + \mathbb{E}\big[\Omega(z, \hat{a})\big] \Big\}. \tag{2.7}$$

As is standard in models built on Lagos–Wright, the portfolio choice is independent of the current state and W is linear in (m, a). Linearity implies that a buyer who enters the next centralized market with real balances z' and assets a' obtains $z' + d\,a'$ up to a constant. Two numbers follow, and every decision below is weighed against them: a unit of real balances the buyer does not spend is worth β to her, and a unit of the asset she carries into the next centralized market is worth βd.

A firm's centralized-market value function is linear for the same reason. Since $i > 0$ a firm carries no money out of the market, so that $W^F(m) = \varphi m + \Upsilon^F$; it carries no assets either.[2]

[2]As Geromichalos and Herrenbrueck (2016) note in the same context, there are equilibria in which the asset is priced at its fundamental value and the firm is indifferent between holding none of it and holding some. A fixed cost of participating in the asset market, however small, makes zero the unique optimum.

The decentralized market

After leaving the centralized market a buyer samples price quotes. With probability α_0 she samples none and will trade no goods this period; otherwise she holds an effective price ρ, the lower of the quotes she drew. Let $\Omega(z, a; \rho)$ and $\Omega_0(z, a)$ denote her value on entering the over-the-counter round in the two cases. A buyer's effective price is at most ρ if she drew one quote below ρ, or two, so the measure of buyers whose effective price is ρ is $\alpha_1 + 2\alpha_2\big(1 - F(\rho)\big)$ and

$$\mathbb{E}\big[\Omega(z, \hat{a})\big] = \alpha_0 \underbrace{\Omega_0(z, \hat{a})}_{\text{lender}} + \int \big[\alpha_1 + 2\alpha_2\big(1 - F(\rho)\big)\big] \underbrace{\Omega(z, \hat{a}; \rho)}_{\text{borrower}} dF(\rho). \tag{2.8}$$

The lender (the buyer who holds no quote). Take this case first, because it is linear and because its solution restricts the price at which the asset can trade. She has no use for money this period. She supplies χ_N units of it in exchange for χ_N/ψ_C units of the asset and proceeds directly to the next centralized market, so that

$$\begin{aligned} \Omega_0(z, a) &= \max_{0 \le \chi_N \le z} \beta \left[\big(z - \chi_N\big) + d \left(a + \frac{\chi_N}{\psi_C} \right) \right] \\ &= \beta\big(z + d\,a\big) + \max_{0 \le \chi_N \le z} g\,\chi_N. \end{aligned} \tag{2.9}$$

What she is weighing is worth setting out slowly, because the entire supply side of the asset market rests on it. Her money is dead weight in this period's goods market: she drew no quote, so there is nothing she can buy with it. It is not worthless, however. Carried into the next centralized market it relieves her of labor one for one, which is why a unit of it is worth β to her in (2.6). Her question is only whether the asset market offers her something better for the one sub-period in which the money would otherwise sit idle.

The rate at which it does so is the one she computes next. One unit of money buys $1/\psi_C$ units of the asset, and each unit of the asset is a claim to d units of the general good in the next centralized market. A unit of money lent therefore turns into d/ψ_C units of next-period goods, against the single unit it turns into if she simply carries it; in present value the two are $\beta d/\psi_C$ and β. Her net gain per unit supplied is the difference between them, a constant we shall use throughout: writing $\kappa \equiv \beta d/\psi_C$ for the value of a unit of money lent, the wedge is $g \equiv \kappa - \beta$, and it is already displayed in the second form of (2.9). Two things about this comparison deserve emphasis. It is between the asset's price and the asset's own dividend, and nothing else enters it: she lends when the asset sells at a discount to what it pays, $\psi_C < d$, because a unit of money then buys more than a unit of goods, and refuses when it sells at a premium. And it is a comparison she makes once rather than unit by unit. Neither side bends — money relieves labor one for one however much of it she carries, and, since she is small and takes ψ_C as given, the hundredth unit of the asset is worth exactly what the first was — so with no curvature anywhere there is nothing to equate at the margin, and whatever settles the first unit settles every unit.

The second form of (2.9) makes this formal: a constant plus a straight line in the only thing she chooses. A linear objective on an interval is maximized at an endpoint, so

$$\chi_N = \begin{cases} z, & \psi_C < d, \\ [\,0, z\,], & \psi_C = d, \\ 0, & \psi_C > d. \end{cases} \tag{2.10}$$

Three features of (2.10) will be used repeatedly. First, the supply of cash to the asset market is $\alpha_0 z$ at most: it is made of the balances of buyers who happened to find no seller, so its size is set jointly by the search technology and by whatever monetary policy does to z. It is not a primitive. Second, the middle line is a correspondence rather than a function. At $\psi_C = d$ the lender is exactly indifferent and any quantity between nothing and everything is optimal for her, so her *individual* supply is not pinned down — though, as Section 2.3 shows, the aggregate still is, by the borrowers on the other side. Third, quantity and price cannot both adjust. Away from $\psi_C = d$ the quantity is stuck at a corner and only the price can move; at $\psi_C = d$ the price is stuck and only the quantity can. Every equilibrium in this paper is of one type or the other, and Proposition 1 is the statement of which arises when.

The third line of (2.10) we may now set aside, and it is worth saying why the case disappears rather than merely being ruled out. At $\psi_C > d$ every lender supplies nothing. There is then no cash on the supply side at all, so no buyer holding a quote can raise any, and the asset market has zero volume at that price whatever the demand for cash happens to be. A price at which nobody trades does not clear the market in any meaningful sense: it is not an alternative regime with its own equilibrium to describe, but a price the market never reaches. Something else would go wrong if it did. The three cutoffs of the next subsection are ordered by the sign of g, and at $g < 0$ the ordering reverses, so a buyer would pass through her financing margins in the opposite sequence and demand would have a different shape entirely. We never have to deal with that shape, because the lender's willingness to trade already confines us to $\psi_C \leq d$, equivalently $g \geq 0$ and $\kappa \geq \beta$, and that is where we stay for the rest of the paper. The restriction is not an assumption imposed for convenience but the range over which one side of the market is prepared to show up.

The borrower (the buyer who holds a quote). A buyer who holds a quote solves the asset market and the goods market as one problem, because the cash she raises in the first is exactly what relaxes her constraint in the second. She may sell assets but does not buy them. Writing χ for the units of money she raises, so that she parts with χ/ψ_C units of the asset,

$$\Omega(z, a; \rho) = \max_{q,\chi} \left\{ u(q) + \beta \left[\left(z + \chi - \rho q \right) + d \left(a - \frac{\chi}{\psi_C} \right) \right] \right\} \tag{2.11}$$

subject to

$$\rho q \leq z + \chi, \qquad 0 \leq \chi \leq \psi_C a. \tag{2.12}$$

The first constraint says she cannot spend more than the cash she ends the asset market with; the second says she cannot raise more than her asset position is worth. A firm that sells q at the price ρ obtains $(\rho - c)q$, which it carries into the centralized market.

Her problem is easier to read once the constant is stripped out of it. Collecting terms in (2.11), the objective is $u(q) - \beta \rho q - g\,\chi$: she pays β for every unit of her own money she spends and a further g for every unit she raises outside those balances, which is why the lender's wedge governs her behaviour too. Write $X \equiv z + \psi_C a$ for her total liquidity, the most she can spend once every unit of the asset has been sold. What remains is to locate the marginal value of cash, $u'(q)/\rho$, against the two thresholds β and κ: she spends all of her own money only above the first and sells assets only above the second. Both comparisons are settled by a single cutoff price each, since $u'(q)/\rho$ falls as ρ rises, so the price line is cut into intervals, one per financing margin — as many of them as the thresholds are distinct, which is to say as the sign of g dictates. The following lemma gives her optimal demand and her use of the asset market in the two cases.

Lemma 1 (Demand)**.** A buyer's optimal decentralized-market behaviour takes one of two forms, according to whether the asset market operates. In each, $q(\rho)$ is the quantity she buys at the quote ρ and $\chi(\rho)$ the cash she raises to pay for it.

(i) *(The asset market operates.)* If $g > 0$, the solution to (2.11) is

$$q(\rho) = \begin{cases} X/\rho, & \rho < \breve{\rho}, \\ (\kappa\rho)^{-1/\sigma}, & \breve{\rho} \le \rho < \tilde{\rho}, \\ z/\rho, & \tilde{\rho} \le \rho < \hat{\rho}, \\ (\beta\rho)^{-1/\sigma}, & \rho \ge \hat{\rho}, \end{cases} \qquad \chi(\rho) = \begin{cases} \psi_C a, & \rho < \breve{\rho}, \\ \rho(\kappa\rho)^{-1/\sigma} - z, & \breve{\rho} \le \rho < \tilde{\rho}, \\ 0, & \rho \ge \tilde{\rho}, \end{cases}$$

where

$$\breve{\rho} = \kappa^{\frac{1}{\sigma-1}} X^{\frac{\sigma}{\sigma-1}}, \qquad \tilde{\rho} = \kappa^{\frac{1}{\sigma-1}} z^{\frac{\sigma}{\sigma-1}}, \qquad \hat{\rho} = \beta^{\frac{1}{\sigma-1}} z^{\frac{\sigma}{\sigma-1}}, \tag{2.13}$$

and these satisfy $\breve{\rho} < \tilde{\rho} < \hat{\rho}$, so that the four branches are the four financing margins in order.

(ii) *(The asset market is idle.)* If $g = 0$ the cutoffs $\tilde{\rho}$ and $\hat{\rho}$ coincide, region III is empty, and regions II and IV carry the same expression. Demand collapses to

$$q(\rho) = \begin{cases} X/\rho, & \rho < \breve{\rho}, \\ (\beta\rho)^{-1/\sigma}, & \rho \ge \breve{\rho}, \end{cases} \qquad \chi(\rho) = \begin{cases} \psi_C a, & \rho < \breve{\rho}, \\ \rho(\beta\rho)^{-1/\sigma} - z, & \breve{\rho} \le \rho < \hat{\rho}, \\ 0, & \rho \ge \hat{\rho}, \end{cases}$$

with the cutoffs of (2.13) evaluated at $\kappa = \beta$. These are the schedules of Head et al. (2012) with total liquidity X in the place of real balances.

Moreover $g > 0 \iff \psi_C < d \iff \breve{\rho} < \tilde{\rho} < \hat{\rho}$, and $g < 0$ is not an equilibrium.

The central feature of part (i) is that only some buyers use the asset market, and which ones is decided by the price they happened to draw. Read the four regions from the right, which is the order in which a buyer meets her financing margins as her quote falls. In region IV the quote is high, she wants little, and her own balances more than cover it: she is unconstrained, buys the frictionless quantity $(\beta\rho)^{-1/\sigma}$, and raises nothing. In region III she has become constrained — she spends every unit of her own money, $\rho q = z$ — and yet she still raises nothing. In region II she does sell, but only as much as she needs, stopping where $u'(q) = \kappa\rho$. In region I even liquidating her entire position leaves her short, so she sells all of it and spends her total liquidity X.

Region III is the one that deserves comment, because it is where a buyer who would like more cash chooses not to obtain it. Her marginal value of cash, $u'(q)/\rho$, has risen above β — that

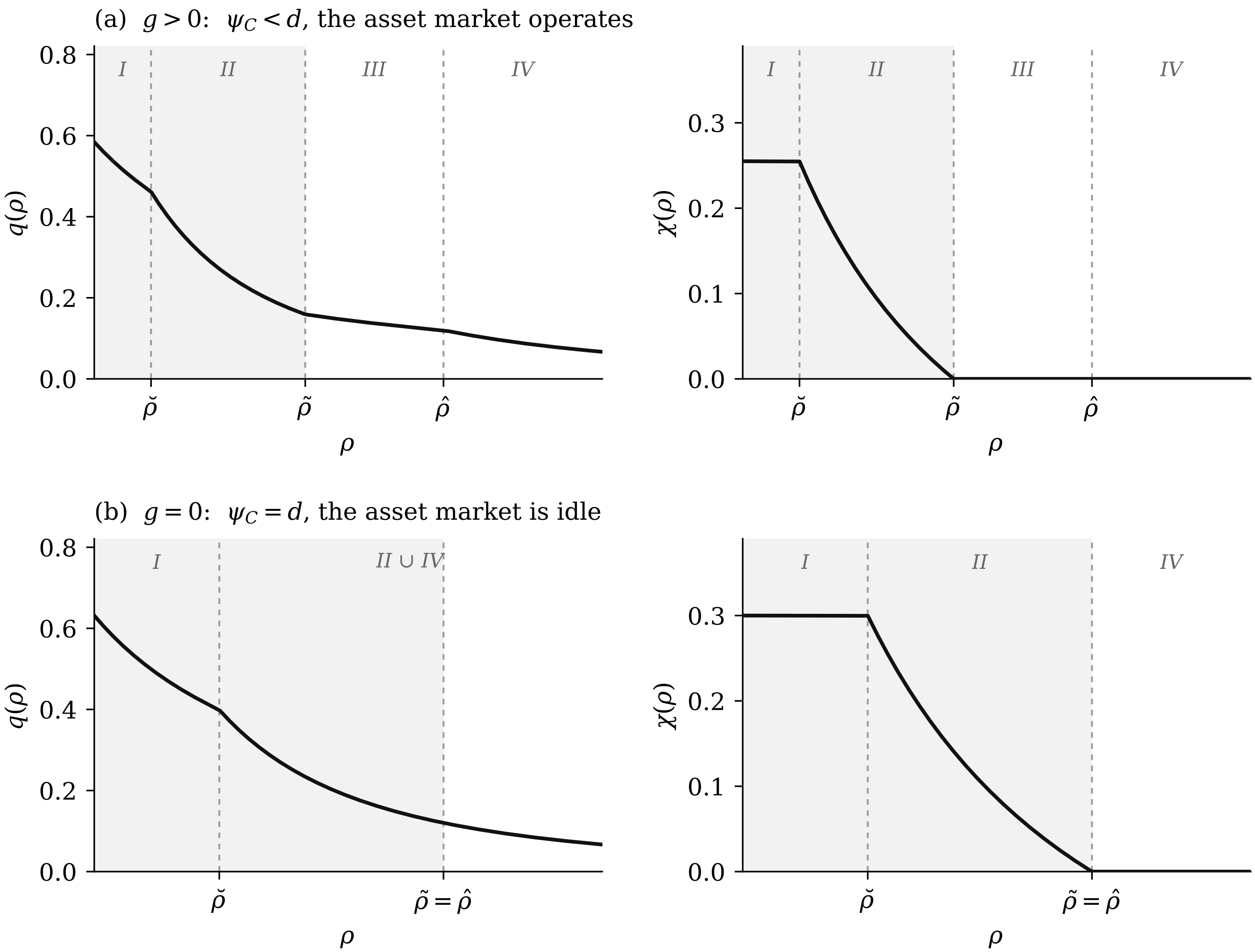


Figure 2: The buyer's demand and her use of the asset market, in the two cases of Lemma 1.

is what makes her spend her own balances down to the last unit — but it has not yet risen above κ. Converting an asset into money costs her g per unit, and at her quote the extra consumption is not worth that price. She is constrained and abstaining at the same time, and the region exists only because $g > 0$. This is what separates the model from Geromichalos and Herrenbrueck (2016), where an exogenous shock sorts buyers into those who need liquidity and those who do not: here the sorting is done by the distribution of posted prices, and it produces three groups rather than two — the unconstrained, the constrained who abstain, and the constrained who trade.

The top row of Figure 2 shows this. In the left panel q falls throughout but with three kinks, one at each cutoff, since the buyer switches financing margin at each and demand is continuous but not smooth there. The right panel is the more informative one. It is flat at zero above $\tilde{\rho}$, covering regions III and IV together, so the two regions that look different in the quantity panel are indistinguishable in the financing panel; it rises as the quote falls through region II, since a better bargain is worth more cash; and it is flat at $\psi_C a$ throughout region I, where the collateral constraint binds. Shading marks the prices at which the asset market is used, and its left edge is $\tilde{\rho}$, not $\hat{\rho}$: the set of buyers who trade is strictly smaller than the set who are constrained.

Part (ii) removes exactly the feature that made region III possible. When $g = 0$ conversion is free, so no constrained buyer has any reason to abstain, and the region between $\tilde{\rho}$ and $\hat{\rho}$ closes up — the two cutoffs are the same number. Regions II and IV, which differed only in whether the marginal unit of cash cost κ or β, become the same problem when the two prices coincide, and they merge into a single branch. What is left is a buyer who cares about her *total* liquidity $X = z + \psi_C a$ and not about how it is divided between money and assets, which is what $\psi_C = d$ means: the two are perfect substitutes at the rate d. Section 3.1 shows that the decentralized market is then identical to one with no asset market at all, so this case is not a curiosity but the one that obtains whenever the asset supply is small.

The bottom row of Figure 2 makes the collapse visible, and also shows what does *not* collapse. The quantity panel now has a single kink instead of three, because there is only one margin left to switch at. The financing panel, however, still changes character at $\hat{\rho}$: χ is positive below it and zero above. The merge is of the quantity, not of the decision to trade. A buyer above $\hat{\rho}$ can already afford what she wants and so has no reason to visit the asset market, while a buyer below it must raise cash even though the terms are now free — which is why the left panel of that row is labelled II $\cup$ IV and the right panel is not.

2.3 The competitive over-the-counter market

The asset market described here is *competitive*: agents take the price ψ_C as given, the market convenes and clears once, and there is neither intermediation nor bargaining. It is the Walrasian counterpart of the over-the-counter market of Geromichalos and Herrenbrueck (2016), where the same asset changes hands in bilateral meetings with search and bargaining in the manner of Duffie et al. (2005). We thus abstract from delay, from meeting frictions, and from the dispersion of bilateral terms of trade, and we retain a single price. Firms stay out, since by Section 2.2 they carry neither money nor assets out of the centralized market.

Both sides of the market are buyers of the decentralized good, and what separates them is the quote each has drawn. Because the market opens *after* quotes have been drawn – the counterpart of the assumption in Geromichalos and Herrenbrueck (2016) that the secondary market convenes after the liquidity shock – a buyer holding a quote knows the price she faces and therefore knows exactly how much cash she is short; she sells assets to obtain it. A buyer who drew no quote will not trade goods this period, so her money is idle until the next centralized market; she buys assets with it in order to collect the dividend. Since the asset is one-period and riskless, selling it outright at ψ_C and pledging it against a loan repaid out of the dividend are the same contract, so that although we write the trade as a sale, the restriction $\chi \leq \psi_C a$ is a collateral constraint in the sense of Ferraris and Watanabe (2008).

Supply is (2.10) and demand is the $\chi(\rho)$ of Lemma 1, aggregated over quotes. Two comparative statics of that demand will matter later. First, χ is non-increasing in ρ: a bargain is worth financing, and the buyers who lean hardest on the asset market are those who found the lowest prices. Second, χ is increasing in ψ_C on every branch, since a higher price raises the value of the collateral, lowers κ, and pushes $\tilde{\rho}$ out. Both the intensive margin and the set of participating quotes expand together.

That last point is where this model departs from the competitive secondary market of Geromichalos and Herrenbrueck (2016). There every agent hit by the liquidity shock is short of cash by the same amount and every one of them trades. Here buyers are hit by the same shock but draw different prices, so at a given ψ_C some are constrained and some are not, and the mass that comes to the asset market is itself an equilibrium object. Price dispersion turns the regime a representative agent would be in into a distribution across agents within the period.

The wedge g is at once the borrower's marginal cost of outside cash and the lender's marginal gain from supplying it, since both face the same price. At the margin the asset round is thus a pure transfer between buyers. It is nevertheless not neutral: the cash moves from someone who cannot use it this period to someone whose constraint binds, so the transfer buys real output. This is the reallocation a bank performs in Berentsen et al. (2007), and the one through which Kam et al. (2024) obtain their externality; here it is performed by an asset market, and what limits a borrower is the value of her collateral rather than the terms of a deposit contract. This is also where the externality of the title enters. A buyer who carries an extra unit of money into the period reduces the aggregate demand for cash in the asset market and so raises ψ_C, which lowers κ for every borrower and lowers the return of every lender. She takes ψ_C as given and internalizes neither effect.

It remains to clear the market. The supply of assets in the centralized market is fixed at A throughout this section and the next, so that $\hat{a} = A$; Section 4 makes it a choice. Writing S for the aggregate supply of cash and aggregating the demand $\chi(\rho)$ of Lemma 1 across quotes, the asset market clears when

$$S = \int \big[\alpha_1 + 2\alpha_2\big(1 - F(\rho)\big)\big]\, \chi(\rho)\, dF(\rho). \tag{2.14}$$

The weight inside the integral is the one that appeared in (2.8): it is the measure of buyers whose effective price is ρ, so (2.14) adds up the cash raised by every buyer who comes to the market, quote by quote.

Clearing alone does not pin the price down, because supply is a correspondence rather than a function. The conditions that do are those carried over from (2.10),

$$\psi_C \le d, \qquad S \le \alpha_0 z, \qquad \big(d - \psi_C\big)\big(\alpha_0 z - S\big) = 0. \tag{2.15}$$

This is where the observation of Section 2.2 — that quantity and price cannot both adjust — is collected. If the second inequality is slack, so that lenders have cash left over, the third condition forces $\psi_C = d$: they are indifferent, and it is the quantity that adjusts to meet whatever the borrowers demand. If instead $\psi_C < d$, the third condition forces $S = \alpha_0 z$: every lender supplies her whole balance, the quantity is at its corner, and it is the price that adjusts. The two cannot both slacken, and which case obtains is a question about how much collateral there is to absorb the idle cash rather than about anyone's preferences.

One feature of (2.14) should be flagged before we use it. Its right-hand side contains F, and F will be shown in Section 2.4 to depend on κ, hence on ψ_C itself. The price in the asset market and the distribution of prices in the goods market are therefore determined together rather than in sequence, which is the sense in which this model differs from one where the secondary market can be solved first and the goods market afterwards. Section 2.5 states the system they jointly satisfy.

2.4 The firm's optimization

A firm posts against a distribution rather than against a price. It commits to a single ρ before any buyer has searched, it cannot see which buyers will sample it, and what it knows is the distribution F from which its rivals' prices are drawn. Its problem is therefore to choose a point on that distribution, and the equilibrium object is a fixed point: each firm takes F as given, and F is what the firms collectively generate.

What a posted price buys is a measure of customers, made of two groups, only the second of which depends on what other firms did. A buyer who sampled this firm and no other buys here whatever the price, and there are α_1 of them. A buyer who sampled two firms buys here only if the other quote was higher, and there are $2\alpha_2(1-F(\rho))$ of them. No third group needs accounting for: two firms post the same price with probability zero, because the F we construct below is continuous.

Profit is the product of two margins, and trading one against the other is the whole of the posting problem. Writing $R(\rho) \equiv q(\rho)(\rho - c)$ for the profit a firm earns from a single customer, given the demand of Lemma 1,

$$\Pi(\rho) = \underbrace{\alpha_1 + 2\alpha_2\big(1 - F(\rho)\big)}_{\text{extensive margin}} \cdot \underbrace{R(\rho)}_{\text{intensive margin}} . \tag{2.16}$$

We write $\varrho(\rho)$ for the first factor and carry it through the rest of the paper: it is the measure of buyers who end up trading at ρ, it falls as ρ rises, and it weights every integral below. Raising the posted price moves the two in opposite directions: it drives buyers to the rival they also sampled, which shrinks the extensive margin, and it widens the mark-up on those who remain, which over most of the range raises the intensive one. The firm chooses where on the price line to stand; the extensive margin is handed to it by everyone else, through F.

A hypothetical monopolist. The top of the support has to be found before anything else, because it is the one price whose profit does not depend on F. A firm posting the highest price is never undercut and never sampled alongside a lower quote, so it sells only to the α_1 buyers who saw it alone and its problem contains no competitor: it maximizes R outright. Write

$$\rho^m \equiv \arg\max_{\rho \geq c} R(\rho) \tag{2.17}$$

for that price. Finding it is not routine: because demand is piecewise, R is not concave and its maximum is not located by setting a derivative to zero. The next lemma solves the problem.

Lemma 2 (The monopolist's price)**.** Let $g \geq 0$ and let $\rho^* \equiv c/(1-\sigma)$ denote the price a monopolist would charge a buyer who is not liquidity constrained. Profit per customer is continuous, strictly increasing on $[c, \breve{\rho})$ and on $[\tilde{\rho}, \hat{\rho})$, and single-peaked at ρ^* on $[\breve{\rho}, \tilde{\rho})$ and on $[\hat{\rho}, \infty)$, the cutoffs being those of (2.13), so R has two humps, one on either side of $\tilde{\rho}$. A local maximum can therefore occur only where a rising branch meets a falling one, which leaves the two kinks and the common peak:

$$\rho^m = \arg\max_{\rho \in \{\breve{\rho},\, \rho^*,\, \hat{\rho}\}} R(\rho), \qquad R(\breve{\rho}) = X\Big(1 - \frac{c}{\breve{\rho}}\Big), \qquad R(\hat{\rho}) = z\Big(1 - \frac{c}{\hat{\rho}}\Big). \tag{2.18}$$

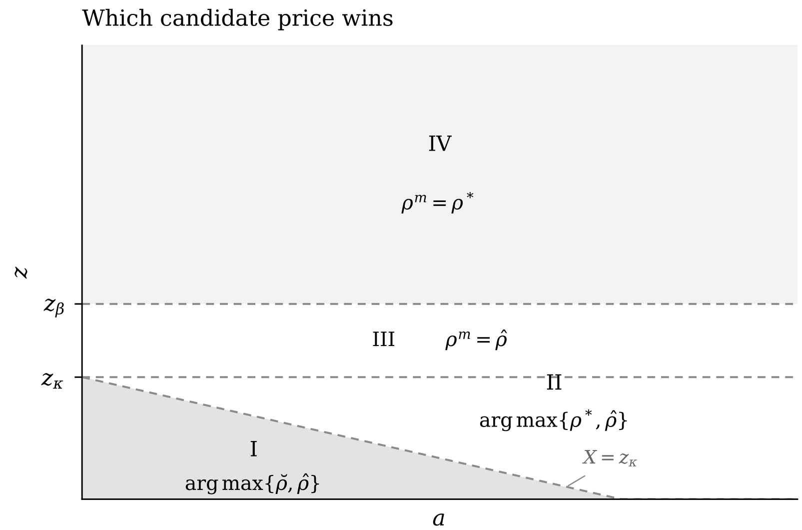


Figure 3: Which of the three candidate prices wins, in the portfolio plane. The label is the region of Lemma 1 in which ρ^* falls.

The three enter symmetrically, and each is the price that just exhausts one notion of purchasing power: $\breve{\rho}$ exhausts *total* liquidity $X = z + \psi_C a$, $\hat{\rho}$ exhausts *cash* alone, and ρ^* exhausts nothing because the buyer charged it is not constrained at all. No side condition is needed in (2.18): a candidate that is not a local maximum is dominated by one that is, so including it changes nothing. If $\breve{\rho} \leq \rho^*$ then R rises through regions I and II and $\breve{\rho}$ loses on its own; if $\breve{\rho} > \rho^*$ then ρ^* lies inside region I and it is ρ^* that loses.

Which of the three survives is decided by the portfolio, and only through which region of Lemma 1 contains ρ^*. Write $z_\beta \equiv \beta^{-1/\sigma}(\rho^*)^{\frac{\sigma-1}{\sigma}}$ and $z_\kappa \equiv \kappa^{-1/\sigma}(\rho^*)^{\frac{\sigma-1}{\sigma}}$ for the balance that just buys the unconstrained quantity at ρ^*, when the marginal unit of cash is the buyer's own money and when it is raised by selling the asset; since $\kappa \geq \beta$, $z_\kappa \leq z_\beta$. Exactly one of the following holds:

region I. $X < z_\kappa$. The buyer charged ρ^* would be short even after liquidating her whole position, so the peak is out of reach and $\rho^m = \arg\max_{\rho \in \{\breve{\rho}, \hat{\rho}\}} R(\rho)$.

region II. $z < z_\kappa \leq X$. She reaches the peak by selling part of her position, so $\rho^m = \arg\max_{\rho \in \{\rho^*, \hat{\rho}\}} R(\rho)$.

region III. $z_\kappa \leq z < z_\beta$. She reaches it out of her own money but only by spending all of it, so $\rho^m = \hat{\rho}$.

region IV. $z \geq z_\beta$. She is not constrained at the peak, so $\rho^m = \rho^*$.

Regions III and IV are horizontal bands in z and do not involve a; regions I and II are separated by the line $X = z + \psi_C a = z_\kappa$ in the portfolio plane. Only in region I does $\breve{\rho}$ win, and only there does the asset supply touch ρ^m.

Grouping the three candidates by the cost of the marginal unit of cash recovers the two humps,

$$\rho_\kappa = \min\big\{\max\{\rho^*, \breve{\rho}\},\ \tilde{\rho}\big\}, \qquad \rho_\beta = \max\{\rho^*, \hat{\rho}\}, \tag{2.19}$$

where ρ_κ is the best price to charge a buyer who is selling assets at the margin and ρ_β the best price to charge one who is not. The two are compared through the values above together with

$$R(\rho^*) = \Phi\,\kappa^{-1/\sigma}\ \text{ in region II}, \qquad R(\rho^*) = \Phi\,\beta^{-1/\sigma}\ \text{ in region IV},$$

according to whether the buyer who is charged ρ^* finances her marginal unit by selling the asset or out of her own money, where $\Phi \equiv \sigma\, c^{(\sigma-1)/\sigma}(1-\sigma)^{(1-\sigma)/\sigma}$ is a constant built from preferences and cost alone. When $g = 0$ the two thresholds coincide, $z_\kappa = z_\beta$, region III is empty, the two humps merge into one, and (2.18) collapses to $\rho^m = \max\{\rho^*, \breve{\rho}\}$.

The two single-peaked branches peak at the same price, and this is not a coincidence: demand has elasticity $1/\sigma$ on both, so κ and β move the level of R but not the location of its peak. Two increasing branches are therefore interleaved with two single-peaked ones, R can carry two local maxima, and the monopolist must compare them — which is the same feature, seen from the seller's side, that produced the fall in Figure 5.

Written this way the departure from Head et al. (2012) is exactly one term. Their monopolist faces a single hump and posts $\max\{\rho^*, \hat{\rho}\}$, which is our ρ_β; the asset market adds a second hump with its own best price ρ_κ, and the highest posted price is settled by comparing *profits* rather than prices. That is why ρ^m can be non-monotone here and is not there, and it is why the maximum in (2.18) cannot be pushed inside R.

It also shows where the asset can act at all. The portfolio enters (2.18) only through $\breve{\rho}$, and $\breve{\rho}$ binds the projection only when $\breve{\rho} > \rho^*$, which is region I. In regions III and IV the buyer clears the bar out of her own money and the seller never looks at her asset position; in region II she holds enough of the asset that ρ^* is already inside the hump and more of it changes nothing. *More of the asset moves the highest posted price only while the buyer is short even after liquidating everything* — which is precisely the flat stretch and the falling stretch of panel (c).

The three candidates have a common reading. When $\rho^m = \breve{\rho}$ the monopolist is targeting buyers who have liquidated their entire asset position, charging the most they can afford. When

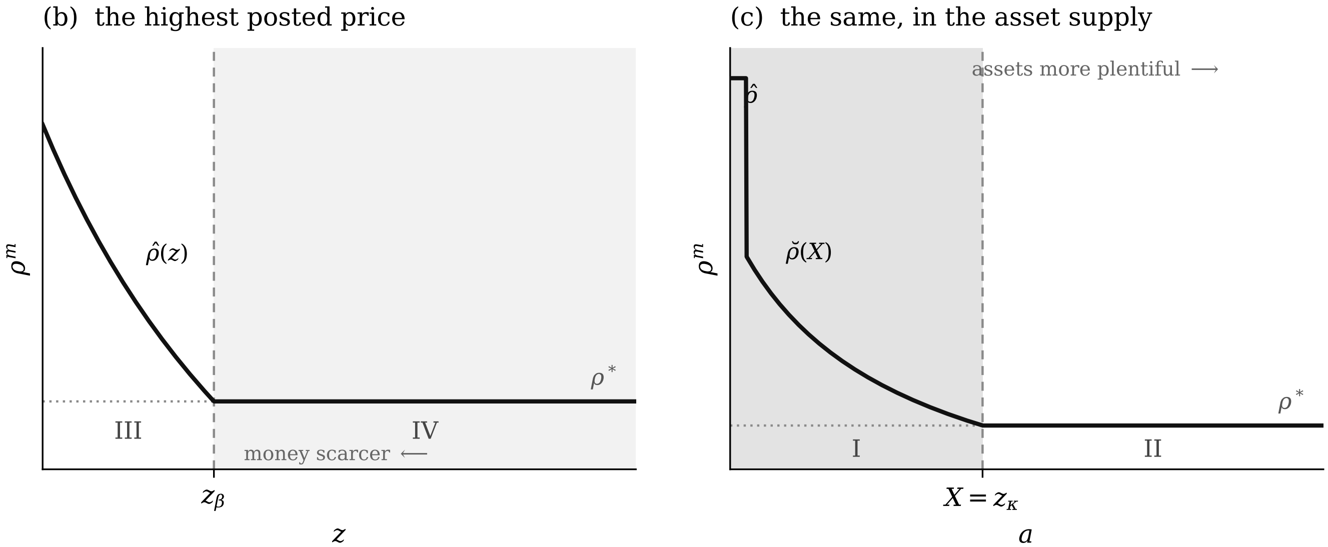


Figure 4: The highest posted price, in money and in assets. Each panel is one axis of Figure 3, and the label on a segment names which of $\breve{\rho}$, ρ^*, $\hat{\rho}$ is the winner of (2.18) there.

$\rho^m = \hat{\rho}$ she targets buyers who are constrained by their own money and stay out of the asset market. When $\rho^m = \rho^*$ she targets unconstrained buyers at the standard constant-elasticity markup.

The list stops at three, although the grouping (2.19) appears to allow a fourth: $\rho_\kappa = \tilde{\rho}$ whenever $\rho^* > \tilde{\rho}$, so that the lower hump is increasing throughout and its best price is its own right endpoint. That candidate never wins. Profit per customer is continuous at $\tilde{\rho}$ and strictly increasing to its right, so $R(\rho_\beta) \geq R(\hat{\rho}) > R(\tilde{\rho})$, the first inequality holding because ρ^* is the peak of the branch on which $\hat{\rho}$ sits. The price at which the buyer stops selling is therefore never the highest price posted: a seller who has raised her price to the point where her marginal customer quits the asset market always does better by raising it further, since that customer is about to spend her cash down to the last unit and has nothing left to withhold. This is why $\tilde{\rho}$ is absent from (2.18) while $\breve{\rho}$ and $\hat{\rho}$ are both in it.

The classification is simpler than it looks: it asks only which region of Lemma 1 the unconstrained monopoly price falls in. Region IV means the buyer charged ρ^* is not constrained at all and the seller can charge it; each step down means she is constrained in one more way, and the seller responds by targeting the constraint instead. The thresholds are where ρ^* crosses a cutoff, and they mean the same thing on the two axes of Figure 3: they are the amounts of liquidity a buyer must carry to reach the unconstrained quantity at ρ^*, and they differ only in what her last unit of cash costs her. Above z_β she gets there out of her own money; above z_κ she gets there only by selling assets, which is dearer, so z_κ is the lower bar. What Lemma 2 says is that the seller's price is decided by which of these bars the buyer clears, and that the asset can be used to clear the lower one.

Read the z axis first, holding a fixed; panel (b) of Figure 4 draws it. Above z_β the monopolist charges the textbook markup ρ^* and the portfolio is irrelevant. Below it she does better to raise her price to $\hat{\rho}$ and take the buyer's entire cash balance, and since $\hat{\rho}$ is decreasing in z, *the highest posted price rises as money becomes scarcer*, with a kink at z_β. Money is thus a brake on the top of the price distribution, and inflation releases it.

Now read the a axis, holding z below z_κ; panel (c) draws it. Raising a raises total liquidity X and pushes $\breve{\rho}$ down, and the monopolist switches from taking the buyer's cash balance at $\hat{\rho}$ to taking her whole liquidated position at $\breve{\rho}$: the highest posted price falls steeply. It stops falling at $X = z_\kappa$ — the boundary between regions I and II — and is flat in a thereafter, because past that point ρ^* is the maximizer and ρ^* contains no portfolio at all. *More of the asset lowers the top of the price distribution, but only until total liquidity reaches z_κ, after which it does nothing.* Assets and money therefore push in the same direction here, but not with the same reach: money works on the whole plane, while the asset works only inside the band it can lift the buyer across.

Read together with Figure 3, the two panels say that what the monopolist responds to is liquidity rather than its composition. Money and assets enter through z and through X, and either can carry the economy from the region in which the seller finds it profitable to price a buyer out of her entire balance to the region in which she settles for the textbook markup. Once liquidity is abundant in that sense the top of the distribution is ρ^*, a number built from c and σ alone, and neither more money nor more assets moves it. Section 3.1 works entirely in that region, which is why the support of F ends at the same ρ^* in every economy we compare there; the asset market is left to act on the rest of the distribution.

The equal-profit condition. With ρ^m in hand the rest of the distribution follows from indifference. In a symmetric equilibrium with dispersion every posted price must earn the same amount, since otherwise some firm would move, so

$$\Pi^* = \big[\alpha_1 + 2\alpha_2\big(1 - F(\rho)\big)\big] R(\rho) \qquad \text{for all } \rho \in \operatorname{supp}(F), \tag{2.20}$$

and evaluating this at ρ^m, where the measure served is α_1, fixes the constant at $\Pi^* = \alpha_1 R(\rho^m)$.

The distribution cannot be recovered from (2.20) by dividing through, and this is where the

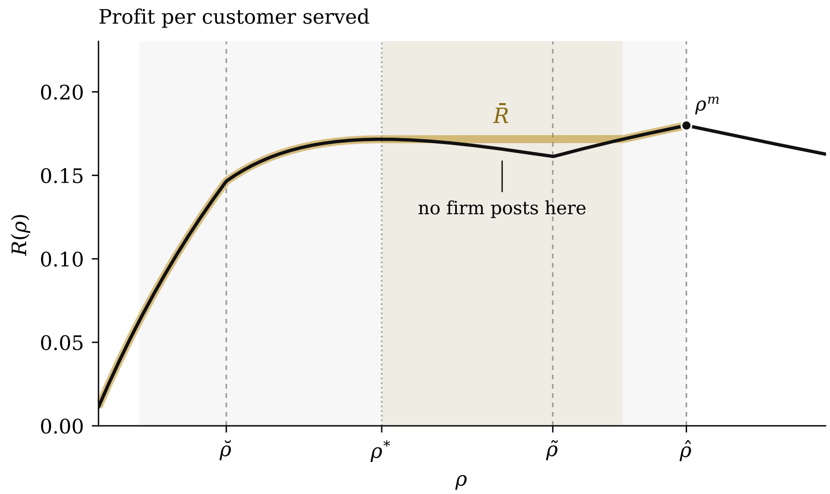


Figure 5: Profit per customer served and its running maximum $\bar{R}$.

model departs from Head et al. (2012). The bracket is non-increasing, because F is a distribution, so R must be non-decreasing wherever F puts mass. In Head et al. (2012) that is automatic: demand has two regions and profit per customer rises throughout. Here it is not. Demand has four regions, and R falls inside the support whenever $\breve{\rho} \le \rho^* < \tilde{\rho}$ and $\rho^m = \hat{\rho}$, a configuration identified in Lemma 2. Dividing through on that stretch returns a *decreasing* F, which is not a distribution.

The asset market is what makes R fall. Past ρ^* a buyer who is financing her purchases by selling assets cuts back sharply as the price rises, because her demand on region II is elastic; the firm loses more on quantity than it gains on the margin. This continues until the price is high enough that she abandons the asset market altogether and spends only her own money, after which her spending is fixed at z and profit per customer rises again. Figure 5 draws such a case: R climbs to a local peak at ρ^*, falls back over the stretch on which region II is being abandoned, and resumes climbing once region III has been reached.

Nothing exotic is needed to repair this, because the falling stretch contains no price a firm would ever post. If some lower price beats ρ on profit per customer, it also serves weakly more buyers, since the measure is non-increasing; it therefore dominates ρ on both margins at once and ρ is never chosen. Prices on the falling stretch simply carry no mass. Writing

$$\bar{R}(\rho) \equiv \max_{c \le \rho' \le \rho} R(\rho') \tag{2.21}$$

for the largest profit per customer available at or below ρ, the distribution follows.

Lemma 3 (The price distribution)**.** Let ρ^m be the monopolist's price of Lemma 2. The equilibrium distribution of posted prices is

$$F(\rho) = 1 - \frac{\alpha_1}{2\alpha_2}\left[\frac{R(\rho^m)}{\bar{R}(\rho)} - 1\right], \tag{2.22}$$

with support $\{\rho : R(\rho) = \bar{R}(\rho)\} \subseteq [\underline{\rho}, \rho^m]$, where the lower bound solves

$$\big(\alpha_1 + 2\alpha_2\big)\bar{R}(\underline{\rho}) = \alpha_1 R(\rho^m). \tag{2.23}$$

F is continuous and carries no atom. When R is non-decreasing on $[\underline{\rho}, \rho^m]$ the support is that whole interval and $\bar{R} = R$.

The proof is omitted. By construction $\bar{R}$ is non-decreasing, so once R is replaced by $\bar{R}$ the equal-profit condition has exactly the form it takes in Head et al. (2012): the bracket in (2.20) and the profit per customer move in opposite directions, the condition can be inverted at every price in the support, and the resulting F is a distribution. Nothing in that argument uses the shape of demand, so it carries over unchanged.

Two properties of (2.22) are worth recording, since Section 3.1 uses both. First, this is an equilibrium: a firm posting ρ earns $\Pi^* R(\rho)/\bar{R}(\rho)$, which equals Π^* on the support and is strictly smaller off it, so no deviation pays. Second, since R is continuous so is $\bar{R}$, and hence so is F — which is what justified dropping the tie term earlier. What the falling stretch produces is therefore a *gap* in the support rather than a jump in F: firms do not post there, but the distribution has no atom at either edge of the gap.

Kam et al. (2024) meet the same non-monotonicity and resolve it differently, and the comparison is worth making because it shows how little machinery the problem needs. In their model the liquidity comes from bank credit rather than from an asset market, and they let firms post lotteries over prices, which replaces R by its concave hull. Enlarging the strategy space in that way makes the replacement necessary: a firm could otherwise reach the hull inside the falling stretch and earn more than Π^*. Under the deterministic posting of Burdett and Judd (1983) and Head et al. (2012) the enlargement never happens, so the weaker repair suffices — the monotone envelope $\bar{R}$ is enough and no randomization enters the model, which leaves the equilibrium object a distribution over prices rather than a distribution over lotteries. The distinction is the one between concavifying a function and merely making it non-decreasing. It is also what separates $\bar{R}$ from the ironing of Myerson (1981), which convexifies the integral of a non-monotone schedule and so pools adjacent types: a mechanism must assign an outcome to every type, whereas a price carrying no mass may simply be omitted.

The classification also settles how the price distribution responds to money, which is the comparative static the mechanism of Section 3 runs on.

Lemma 4 (Price dispersion and real balances)**.** Fix $\psi_C \leq d$, let $z_0 < z_1$, and write $\Lambda(\rho; z) \equiv \bar{R}(\rho; z)/R(\rho^m; z)$, so that (2.22) reads $F = 1 - \frac{\alpha_1}{2\alpha_2}\left[\Lambda^{-1} - 1\right]$ and F is increasing in Λ at every price.

(i) Let $\rho \geq \tilde{\rho}$ be a price that firms post with positive probability in the economy with the smaller balance, evaluating $\tilde{\rho}$ there. Then $\rho^m = \max\{\rho^*, \hat{\rho}\}$ in both economies and

$$\Lambda(\rho; z) = \begin{cases} \Phi^{-1}\beta^{1/\sigma}\, z\,(1 - c/\rho), & \rho^m = \rho^* \text{ and } \tilde{\rho} \leq \rho < \hat{\rho}, \\ \Phi^{-1}\rho^{-1/\sigma}(\rho - c), & \rho^m = \rho^* \text{ and } \rho \geq \hat{\rho}, \\ \dfrac{1 - c/\rho}{1 - c/\hat{\rho}}, & \rho^m = \hat{\rho}. \end{cases} \tag{2.24}$$

Each line is non-decreasing in z — strictly in the first and the third, constant in the second — so $F(\rho; z_0) \leq F(\rho; z_1)$, whichever case of Lemma 2 obtains. Buyers who pay out of their own money face weakly higher prices when real balances are smaller. The second line contains neither z nor ψ_C nor a, and the discount factor has cancelled, so on region IV

$$F(\rho) = 1 - \frac{\alpha_1}{2\alpha_2}\left[\frac{\Phi\,\rho^{1/\sigma}}{\rho - c} - 1\right], \qquad \rho \geq \hat{\rho}, \tag{2.25}$$

which is the same distribution in every economy.

(ii) If $z_0 \geq z_\beta$ then $R(\rho^m; z) = \Phi\beta^{-1/\sigma}$ at both, and $\bar{R}(\rho; \cdot)$ is non-decreasing at every ρ, so $\Lambda(\rho; z_0) \leq \Lambda(\rho; z_1)$ for all ρ: $F(\cdot; z_0)$ first-order stochastically dominates $F(\cdot; z_1)$.

(iii) Otherwise $R(\rho^m; z) = z - c\beta^{\frac{1}{1-\sigma}} z^{\frac{1}{1-\sigma}}$, which rises with z wherever $c\beta^{1/(1-\sigma)} z^{\sigma/(1-\sigma)} < 1 - \sigma$. Numerator and denominator of Λ then move together, and the dominance may fail. Part (i) is unaffected.

Part (i) says that one group of buyers can be tracked without knowing anything about the asset market, and the reason is worth stating because the paper turns on it. A price at the top of the support is the price a monopolist would charge, and by Lemma 2 the customer she has in mind there is either unconstrained or constrained by her own money — never one who is financing at the margin by selling the asset, since $\check{\rho}$ lies below $\tilde{\rho}$. The benchmark against which every other price is judged is therefore built out of β and z alone. Divide the profit a firm earns from a cash-paying customer by that benchmark and the real balance cancels or drops out entirely, leaving a ratio that can only fall as money becomes scarcer. Buyers who never enter the asset market are in this precise sense insulated from it.

The economics of the comparison is about how hard sellers compete, not about what buyers can afford. Equilibrium requires every posted price to earn the same profit, and there are two ways to earn it. A firm can shade its price and win the shoppers who are holding a rival's quote as well; what that victory is worth is the extra revenue those shoppers bring, and that is bounded by what they are carrying. Or it can post high and sell only to the captives who saw no one else, collecting the monopoly profit per customer.

Take real balances away and the two move apart. The return to undercutting falls, because a constrained buyer converts a price cut into fewer extra units; the return to exploiting a captive does not fall as fast, and in region IV does not fall at all. A low price must therefore be compensated with more customers than before, and the only way the market can promise that is to make it less likely that anyone undercut it — which is to shift the distribution up. Poorer buyers are met with higher prices, and the mechanism is that their poverty has blunted the incentive to compete for them.

This is why liquidity belongs in the theory of markups. Inflation here does not merely tax purchasing power at a given price; it slackens the product market, and the markup rises on top of the tax. What parts (ii) and (iii) add is that the statement is uniform only while the monopolist is charging the textbook markup. Once she is pricing a constrained buyer out of her whole balance, more money raises the top of the support as well as the bottom, the two effects work against each other, and the distributions can cross. Section 3.1 shows the asset market producing exactly such a crossing, and part (i) is what survives it.

2.5 Stationary monetary equilibrium

We now construct the stationary monetary equilibrium. Three of its four conditions are already in hand — the demand of Lemma 1, the clearing conditions of Section 2.3, and the price distribution of Lemma 3 — and the fourth, the portfolio choice that ties them together, is what this section supplies. It is also where the asset market shows itself most plainly, because the marginal value of a unit of cash is a different object on each of the four regions of Lemma 1: the equation that prices money carries one term for each of them, and which terms it carries is itself an equilibrium outcome.

Differentiating (2.6) and applying the envelope theorem to (2.11) and (2.9) gives two portfolio conditions. A unit of real balances costs $1+\mu$ to carry and returns its expected marginal value, which is $\kappa = \beta d/\psi_C$ in the state where the buyer draws no quote and $u'(q(\rho))/\rho$ in the state where she draws ρ; a unit of the asset costs ψ and returns the dividend plus whatever constraint it relaxes. Neither condition can be written down without saying which region of Lemma 1 the buyer is in, because $u'(q(\rho))/\rho$ is a different expression on each.

It is worth being clear about what is chosen and what is taken as given, because the two appear in the same integral. A buyer picks her portfolio in the centralized market, before her quote is drawn, so she does not know which region she will land in; what she knows is the distribution it will be drawn from. Her first-order condition therefore averages the four regions with the weights $\varrho\, dF$, which the market hands her, while the limits of integration are her own cutoffs (2.13), which move with the (z, a) she is choosing. The firm likewise takes F as given, but the equal-profit condition of Lemma 3 closes F as a function of (z, ψ_C). So nothing is circular: the following lemma is a first-order condition, and the two market-clearing conditions then pin the pair (z, ψ_C) down.

Lemma 5 (Money demand and asset demand)**.** Real balances and the asset satisfy

$$\frac{1+\mu}{\beta} \;=\; 1+\mathcal{R}(z,\psi_C), \tag{2.26}$$

$$\psi \;=\; \underbrace{\beta d}_{\text{fundamental}} \;+ \underbrace{\psi_C \int_{\underline{\rho}}^{\breve{\rho}} \varrho \left[X^{-\sigma}\rho^{\sigma-1} - \kappa\right] dF}_{\text{liquidity premium}}, \tag{2.27}$$

where the liquidity premium on money is

$$\mathcal{R}(z,\psi_C) = \underbrace{\alpha_0 \frac{g}{\beta}}_{\substack{\text{no quote:}\\ \text{she lends it}}} + \underbrace{\int_{\underline{\rho}}^{\breve{\rho}} \varrho \left[\frac{X^{-\sigma}\rho^{\sigma-1}}{\beta} - 1\right] dF}_{\substack{\text{region I:}\\ \text{sells all, still short}}} + \underbrace{\frac{g}{\beta}\int_{\breve{\rho}}^{\tilde{\rho}} \varrho\, dF}_{\substack{\text{region II:}\\ \text{saves a sale}}} + \underbrace{\int_{\tilde{\rho}}^{\hat{\rho}} \varrho \left[\frac{z^{-\sigma}\rho^{\sigma-1}}{\beta} - 1\right] dF}_{\substack{\text{region III:}\\ \text{spends all her cash}}}. \tag{2.28}$$

There is one term for the buyer who draws no quote and one for each of regions I, II and III of Lemma 1, in the same order and over the same cutoffs; region IV contributes nothing. The premium in (2.27) is $\psi_C\beta$ times the region I integral.

In contrast to Head et al. (2012), the liquidity premium on money in (2.28) now depends on what the buyer can raise in the secondary market, and it comprises four terms. First, with probability α_0 she draws no quote at all and holds cash she cannot spend; she converts it into the asset at ψ_C and collects d on every unit, so her marginal value of a unit of cash is κ and her term is the constant $\alpha_0 g/\beta$. Second, if she draws a price below $\breve{\rho}$ she liquidates her whole position and is still short, spends $X = z + \psi_C a$, and values cash at $X^{-\sigma}\rho^{\sigma-1}$; this is the only term that contains the asset supply, and so the only route by which A reaches money demand at all. Third, if she draws a price in $[\breve{\rho}, \tilde{\rho})$ she sells only what she needs, equates $u'(q)$ to $\kappa\rho$, and so values cash at the constant κ whatever price she drew — the corresponding constant in Head et al. (2012) is β, and the gap between them is g: one letter, and it is the whole of the wedge the asset market drives into money demand. Last, if she draws a price in $[\tilde{\rho}, \hat{\rho})$ she spends all her cash and sells nothing, and values it at $z^{-\sigma}\rho^{\sigma-1}$; nothing in her problem contains ψ_C or a, and she is the buyer Lemma 4(i) is about. A buyer who draws a price above $\hat{\rho}$ is not constrained, values cash at exactly β, and carries no premium: region IV contributes no term, which is why there are four and not five.

The four terms divide in two. On the two regions where the buyer is short her marginal value of cash depends on the price she drew, so the term is a genuine integral; on the two where she is not it is a constant and the integral collapses to a measure. That division is what makes R log-submodular in (ρ, z) and so is what the proof of Lemma 4 runs on.

A term survives exactly when the support of F meets its range, so which terms are present is a statement about (z, a), read through results already in hand. Lemma 2 places the upper end ρ^m in the portfolio plane of Figure 3, and the equal-profit condition of Lemma 3 sets the lower end $\underline{\rho}$. Three readings matter. With no asset, $\breve{\rho}$ and $\tilde{\rho}$ coincide and region II is empty: that term exists only because the asset does. With a great deal of it, X is large, $\breve{\rho}$ has fallen below $\underline{\rho}$ and region I is empty instead; since A enters (2.28) only through X inside that term, the asset supply then leaves (2.26) and (2.14) altogether, the premium in (2.27) vanishes, and $\psi = \beta d$. And when the secondary market prices the asset at par, $\psi_C = d$ and $g = 0$, the other three terms go at once — two carry the factor g and the third is an interval that closes with it — leaving region I alone, which is the Euler equation of Head et al. (2012) with real balances replaced by total liquidity; total liquidity then does not depend on A and $dz/dA = -d$, so the asset displaces money one for one and nothing else changes.

The asset side goes the same way and reduces further still. By Lemma 5 the premium in (2.27) is carried by region I alone, so it too is zero at $\psi_C = d$, where the bracket vanishes, and zero again once that region is empty: $\psi = \beta d$ at both ends. Neither statement is about a choice. Whether $\psi_C = d$, and whether $\breve{\rho}$ has fallen to $\underline{\rho}$, are settled once (z, ψ_C) is solved for, and Proposition 1 answers both in A and μ alone.

It remains to record the two market-clearing conditions and to collect everything. The asset market clears as in Section 2.3: cash supplied by lenders meets cash demanded by borrowers, (2.14), subject to the complementary slackness (2.15) that keeps ψ_C at or below d. On the seller's side, Lemma 2 fixes the top of the support and Lemma 3 then delivers the whole distribution from the equal-profit condition. Nothing further is needed.

Definition 1 (Stationary monetary equilibrium)**.** Given $\mu > \beta - 1$ and A, a stationary monetary equilibrium is a list $\left(z, \psi, \psi_C, S, F, \rho^m\right)$ with $z > 0$ such that

(i) $q(\rho)$ and $\chi(\rho)$ solve the buyer's problem (2.11), as characterized in Lemma 1;

(ii) the portfolio conditions (2.26) and (2.27) hold;

(iii) F is a distribution with support $[\underline{\rho}, \rho^m]$ satisfying the equal-profit condition (2.20), with ρ^m given by Lemma 2;

(iv) the centralized asset market clears, $\hat{a} = A$;

(v) the asset market clears, (2.14) and (2.15);

(vi) real balances are constant, $z = \hat{\varphi}\hat{M}$ with $\varphi/\hat{\varphi} = 1 + \mu$.

The list is long but the system is short. Everything in Definition 1 is a function of the pair (z, ψ_C), and the two conditions that pin that pair down are the money Euler equation (2.26) and the clearing condition (2.14); the asset price ψ then follows from (2.27).

Both hypotheses of the proposition below are statements about how the distribution of prices moves with real balances, and the first is where Lemma 4 earns its keep. Head et al. (2012) and Kam et al. (2024) obtain the monotonicity they need from a first-order stochastic dominance result, which they assume; here part (ii) of Lemma 4 supplies exactly that dominance whenever $z \geq z_\beta$, so on that range the ordering is a result rather than an assumption. What the proposition asks for is in any case weaker: dominance orders $\int h\, dF$ for *every* increasing h, whereas only the single integral in (2.26) is at issue. Both hypotheses are met throughout the range computed in Section 3.[3]

The two collapses of (2.28) say that the equilibrium system is not one system but several. At $\psi_C = d$ it becomes a single equation in the single unknown X, and money and the asset are perfect substitutes; with region I empty it is two equations in (z, ψ_C) from which A has disappeared, so the asset supply is irrelevant and its price is fundamental. Which of the three obtains is settled by A and μ together. Existence, uniqueness and that classification are one result.

[3] Geromichalos and Herrenbrueck (2016) likewise add an assumption to obtain uniqueness, requiring $u'(q)q$ to be non-decreasing so that the cutoff μ_{II} of their Lemma 4 is unique.

Proposition 1 (Stationary monetary equilibrium)**.** Let $\mu > \beta - 1$ and $\lambda \in (0, 1)$, and suppose that raising real balances does not raise the liquidity premium, so that the right-hand side of (2.26) is decreasing in z. Then a stationary monetary equilibrium exists, and it is unique if in addition the demand for cash is increasing in ψ_C. There are cutoffs $\underline{A}(\mu) < \overline{A}(\mu)$ such that it is in exactly one of three regimes:

(i) *idle*, if $A < \underline{A}(\mu)$: the asset market is open but $\psi_C = d$, so only the region I term of (2.28) survives and real balances are $z = \bar{X}(\mu) - dA$;

(ii) *active*, if $\underline{A}(\mu) < A < \overline{A}(\mu)$: $\psi_C < d$, region I carries positive mass, all four terms survive, and $\psi > \beta d$;

(iii) *irrelevant*, if $A > \overline{A}(\mu)$: region I is empty, so the remaining three terms of (2.28) survive without it, the equilibrium does not depend on A, and $\psi = \beta d$.

Both cutoffs are decreasing in μ.

We call these regimes 1, 2 and 3, and use the numbers from here on. The proof of existence and uniqueness is in the appendix; that the two cutoffs exist and fall with inflation is verified numerically over the parameter range of Section 3.

The economics is that of a market whose size is set on both sides at once. Below $\underline{A}$ there is not enough collateral for borrowers to absorb the cash that lenders would gladly supply, so the price cannot fall below d and the asset is priced as money; every extra unit of it displaces exactly d units of real balances and nothing real happens. Above $\overline{A}$ there is so much collateral that no borrower is ever capped by it, and further units are pure dividend. In between, and only in between, does the supply of the asset change anything. Both cutoffs fall with inflation because inflation shrinks z: it lowers the cash lenders bring, which lowers $\underline{A}$, and it raises the number of buyers whose own balances fall short, which lowers $\overline{A}$. Figure 6 draws the partition; Section 3 puts numbers on it and shows that the sign of what the asset does to welfare is not the same throughout.[4]

[4]The counterpart in Geromichalos and Herrenbrueck (2016) is their Lemma 4, which partitions (A, μ) space using cutoffs available in closed form, and their Figure 2. Ours are not closed form, because the integrals in (2.26) depend on F; the classification, however, is the same, and their Regions 5 and 3 correspond to our regimes 1 and 3.

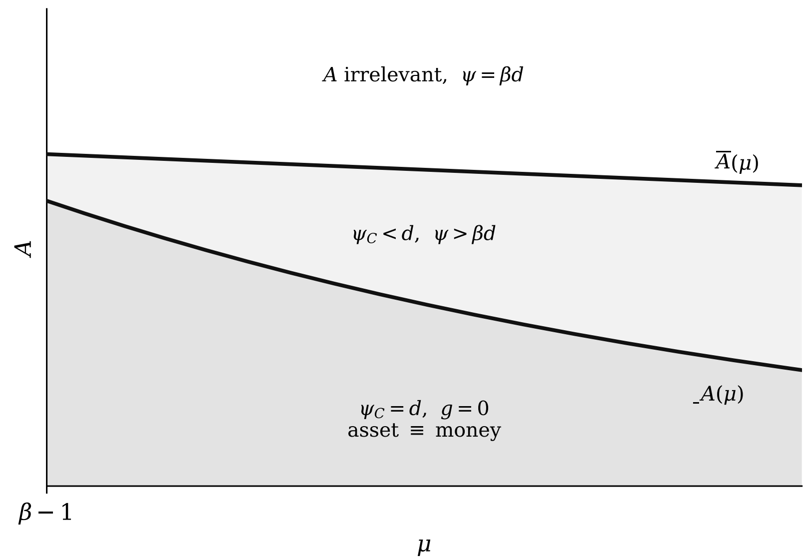


Figure 6: The three regimes of Proposition 1, drawn schematically.

3 Pecuniary externality in the competitive OTC market

This section asks what the asset market does to the distribution of prices at which buyers trade, and what that does to welfare. The question is comparative, so it needs a benchmark, and the natural one is the same economy with the asset market closed.

The benchmark of Head et al. (2012). Let the asset still be issued and still pay d, but let it be impossible to sell in the interval between the centralized and the decentralized market. A buyer then carries it only for its dividend, and the only liquidity she brings to a quote is her own cash. Formally this is the case $\chi \equiv 0$, in which regions I and II of Lemma 1 are empty and κ is replaced by β everywhere. What remains is the economy of Head et al. (2012). We write z_0, F_0, $\underline{\rho}_0$ and ρ_0^m for its equilibrium objects.

The benchmark is therefore not a different model but a restriction of ours, and its four equilibrium objects are obtained from Section 2.5 by deletion. Demand has two regions instead of four,

$$q_0(\rho) = \begin{cases} z_0/\rho, & \rho < \hat{\rho}_0, \\ (\beta\rho)^{-1/\sigma}, & \rho \geq \hat{\rho}_0, \end{cases} \qquad \hat{\rho}_0 = \beta^{\frac{1}{\sigma-1}} z_0^{\frac{\sigma}{\sigma-1}}, \tag{3.1}$$

so that expenditure $\rho q_0(\rho)$ is flat at z_0 below $\hat{\rho}_0$ and falls above it. Money demand is (2.28) with the no-quote, region I and region II terms struck out, which leaves a single integral,

$$i \;=\; \int_{\underline{\rho}_0}^{\hat{\rho}_0} \big[\alpha_1 + 2\alpha_2\big(1 - F_0(\rho)\big)\big] \left[\frac{z_0^{-\sigma}\rho^{\sigma-1}}{\beta} - 1\right] dF_0(\rho). \tag{3.2}$$

The upper limit is the whole of the content. Real balances enter money demand only on $[\underline{\rho}_0, \hat{\rho}_0)$, where the buyer spends all of them; above $\hat{\rho}_0$ she is unconstrained, the bracket is zero, and there is nothing to integrate. This is the form of Head et al. (2012), whose equation (B.12) is the same integral cut off at the same cutoff, and of Kam et al. (2024), whose liquidity premium likewise equals i term by term. The distribution is (2.22) with $\bar{R}$ replaced by R,

$$F_0(\rho) = 1 - \frac{\alpha_1}{2\alpha_2}\left[\frac{R_0(\rho_0^m)}{R_0(\rho)} - 1\right] \quad \text{on} \quad [\underline{\rho}_0, \rho_0^m], \qquad \rho_0^m = \max\{\rho^*, \hat{\rho}_0\}. \tag{3.3}$$

The single parameter κ thus carries the entire difference between the two economies. It enters (2.26) directly, through the value of cash to a buyer who draws no quote, and indirectly, through the two cutoffs $\breve{\rho}$ and $\tilde{\rho}$ that open regions I and II.

3.1 The pecuniary externality in the distribution of prices

The asset market is a gain from trade. A buyer who draws a good quote can turn assets into cash exactly when she wants to spend, and buy more than her own money would allow. But she leaves the centralized market knowing she will be able to, and so she carries less of that money. The buyer whose quote turns out to be poor, and who therefore does not use the asset market at all, is left with the smaller balance and nothing to supplement it. The asset market has two faces, and which of them dominates is not obvious. It is a question about who faces which prices.

Two results already in hand fix the direction. The first is that real balances and the distribution of prices are not independent objects. By Lemma 4(i) a buyer who pays out of her own money faces weakly higher prices when her balance is lower: a poorer buyer converts a price cut into fewer extra units, so the return to undercutting falls relative to the return to exploiting a captive, and the distribution must shift up to keep firms indifferent. That is a result rather than an assumption. The second is that the asset market lowers real balances, and this can be read off the shape of the two Euler equations. The benchmark's (3.2) carries a single term, because the only way to relax the decentralized constraint is to carry cash into the market. Ours, (2.28), carries four, and the three that are new are three ways of obtaining liquidity without carrying cash: idle money earns the dividend, the asset supplements money, and the asset replaces it at the margin. At the same nominal rate the buyer therefore economizes on money.

Put the two together and the direction is this: the asset market lowers real balances, and lower real balances raise the prices faced by the buyer who pays cash. We follow it by comparing two objects with the benchmark, the expenditure schedule and the distribution of posted prices, and we compare them at three asset supplies, because the answer depends on how much of the asset there is. The chain is right in its conclusion and wrong in its first link, and seeing why is the business of what follows: what matters is not that balances fall but that the asset trades below its dividend.

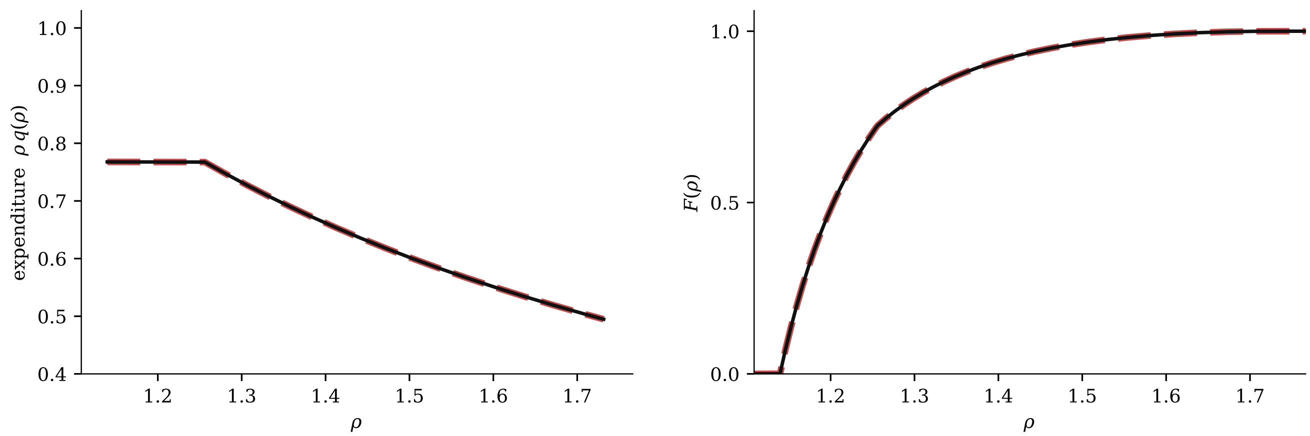
Since the question is how the comparison changes with the quantity of the asset, we hold inflation fixed and raise A.[5] The three rows of Figure 7 are the same economy at three asset supplies, and they are the three regimes of Proposition 1: *scarce*, *intermediate*, *plentiful*. In each row the left panel is the expenditure schedule $\rho\, q(\rho)$, which records what a buyer does with a quote, and the right panel is the distribution $F(\rho)$, which records what quotes she faces. The dashed red curve is the benchmark with the asset market closed and the solid black curve the economy with it open; shading marks the prices at which the asset market delivers cheaper quotes (blue) and dearer ones (pink). Axes are common to all three rows. The two columns are paired for the reason just given: the left is purchasing power and the right is the environment that purchasing power meets, and the answer turns on whether a given buyer is treated the same way by both.

Consider the scarce case, the top row. The two schedules coincide, and they coincide exactly: black and dashed red differ by no more than machine precision, in the distribution as in expenditure. Every object that describes the decentralized market – the support, the mean posted price, the mean transacted price – is the object of the benchmark.

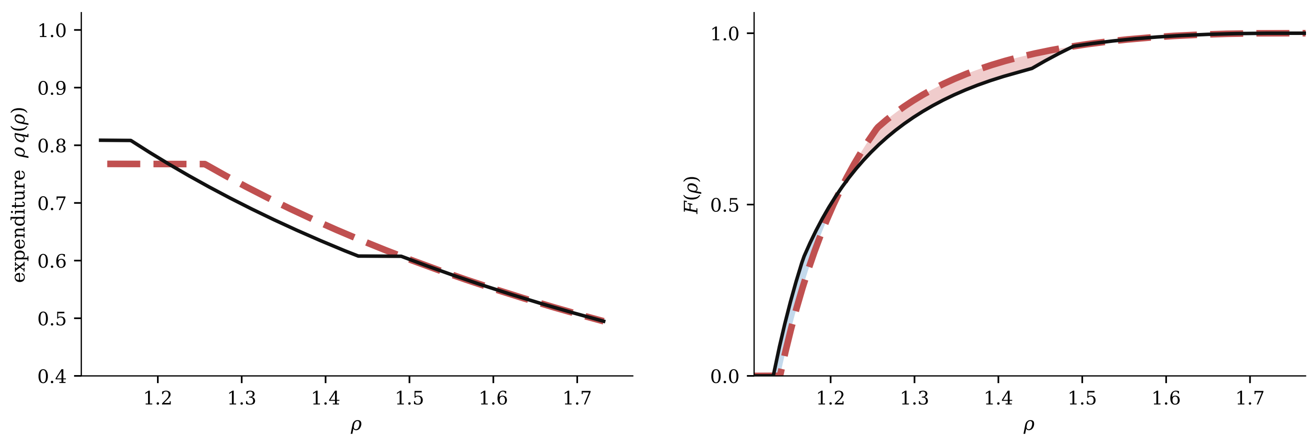
The coincidence is not the statement that nothing happens. The asset is held, it is priced at $\psi_C = d$, and buyers who draw low quotes do sell it. What is true is that real balances fall by exactly the liquidity the asset supplies. Lemma 5 is the algebra: at $\psi_C = d$ we have $\kappa = \beta$, so three of the four terms of (2.28) vanish together and the one that survives depends on the portfolio only through total liquidity. Our four terms have collapsed back to the benchmark's one, with X in place of z; so X solves (3.2), $dz/dA = -d$, the crowding out is complete and the replacement exact. Money and the asset are perfect substitutes, and the composition of the portfolio is indeterminate in everything the decentralized market can see.

This is where the first link of the chain breaks. Balances have fallen, and by more than

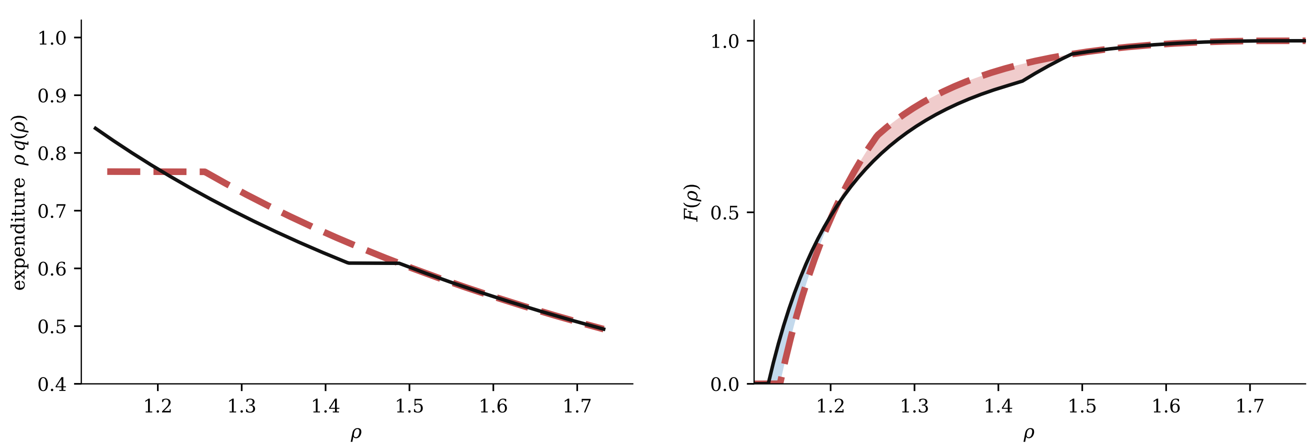
[5] Throughout this section $\beta = 0.9804$, $\sigma = 0.4225$, $c = 1$, $d = 0.10$, $\lambda = 0.55$ and $\mu = 0.30\%$, so that $(\alpha_0, \alpha_1, \alpha_2) = (0.2025, 0.4950, 0.3025)$. The three rows are drawn at $A = 0.85$, $A = 2.05$ and $A = 4.00$.



(a) *Scarce* ($A < \underline{A}$, regime 1). The asset is money: it trades at $\psi_C = d$, displaces real balances one for one, and the two economies coincide exactly.



(b) *Intermediate* ($\underline{A} < A < \overline{A}$, regime 2). The asset is liquidity: it trades below d, regions I to III all carry mass, and the distributions cross.



(c) *Plentiful* ($A > \overline{A}$, regime 3). The asset is a dividend claim: region I is empty and the asset supply has left the equilibrium.

Figure 7: The asset market against the benchmark, from scarce to plentiful.

anywhere else in the figure, and the distribution of prices has not moved at all. A pecuniary externality running through the asset market is not a mechanical consequence of having one. Here is an economy in which buyers hold assets, sell them for cash, and face a distribution of prices, and in which that distribution is nonetheless the one they would face if the asset market did not exist. Something more is needed, and Proposition 1 says what: the asset must be scarce enough on the borrowing side that its price falls below d.

Cross $\underline{A}$ and one thing changes: the market can no longer clear at $\psi_C = d$. Borrowers bring more collateral than lenders will absorb at that price, so ψ_C falls, and with it $\kappa = \beta d/\psi_C$ rises above β. Everything in the rest of this subsection follows from that single movement.

The first consequence is in the left panel of the middle row. The benchmark schedule has one plateau; ours has two. Expenditure is flat at total liquidity over region I, where the buyer has sold her entire position and is still short, and flat again at her own balance over region III, where she has spent all her cash and will not touch the asset market; between them, on region II, she finances at the margin by selling and expenditure slopes down. The four regions of Lemma 1 are visible here as shape. The benchmark cannot produce it, because with one financing margin it has one plateau.

The levels are the crowding out drawn as a picture. At low quotes the black curve lies *above* the red: total liquidity exceeds the benchmark's balance, so a buyer who draws a cheap price buys more than she could without the asset market. Over the middle of the support it lies *below*: her own balances have fallen, and on region III that is all she can spend. Purchasing power has moved from the middle of the price distribution to the bottom.

The second consequence is what firms do about it, and it is in the right panel. Since $\rho^m = \rho^*$ in both economies and the buyer charged ρ^* is unconstrained, the profit of the highest-priced firm is the same number in both: by Lemma 2 the top of the support contains no portfolio at all. The equal-profit condition of Lemma 3 then makes F a strictly increasing function of $\bar{R}$ alone, so the two distributions are ordered at a price exactly as the two profit schedules are. Reading $R = q(\rho)(\rho - c)$ off Lemma 1 region by region settles that ordering: on region I it rises with total liquidity; on region II it falls as κ rises; on region III it falls with the real balance, which is what Lemma 4(i) turns into a statement about prices; and on region IV it does not contain the asset supply at all.

The distributions must therefore cross, and the right panel shows them doing it. At the bottom F lies above F_0 and prices appear that the benchmark never posts. Over the bulk of the support F lies below F_0: those prices are dearer. Above $\hat{\rho}$ the two coincide exactly, and not approximately: by (2.25) the distribution there is a function of the price alone, whatever the real balance, the secondary-market price and the asset supply. The asset market cannot touch the top of the price distribution; what it does is buy a thin cheap tail at the cost of making the middle dearer. The crossing is the failure of dominance anticipated in Lemma 4(iii), produced here by the asset supply rather than by the real balance, and part (i) of that lemma is what survives it.

Two groups are therefore treated in opposite ways, and each of them is treated the same way twice. The buyer who draws a low quote gains on both margins: she has more to spend, and she is quoted less. The buyer who pays out of her own money loses on both: she has less to spend, and she is quoted more. Nothing in her problem contains the secondary-market price or the asset position – she is the buyer of region III, who never enters the asset market at all – and she nonetheless pays the price that market sets. The mean posted price rises, and that is the pecuniary externality. It is exactly zero in the scarce case: it switches on at $\underline{A}$ and not at $A = 0$, so it is not the existence of the asset market that raises prices but the fact that the asset trades below its dividend.

Above $\overline{A}$ one more thing changes, and it is a boundary rather than a price: $\breve{\rho}$ falls below $\underline{\rho}$ and region I empties. No buyer, at any quote in the support, is capped by her collateral. The bottom row records this in the plainest way available: the expenditure schedule never reaches total liquidity. In the intermediate case the curve rested on that level over an interval; here it does not reach it at all, because doing so would require selling the whole position, and no quote is cheap enough to make that worthwhile.

The asset supply thereupon leaves the model. Region I carries the only term of (2.28) that contains A, and the clearing condition over the region that remains does not contain it either, so Proposition 1(iii) is the statement that A drops out of both equilibrium conditions at once. Different asset supplies above $\overline{A}$ deliver the same real balance, the same secondary-market price, and price distributions that agree to machine precision.

The externality is accordingly larger here than in the intermediate case and no longer growing. What orders its size is not the quantity of the asset but its price. Both the wedge $\kappa - \beta$ and the deviation of F from F_0 are governed by how far ψ_C has fallen below d, and ψ_C stops falling at $\overline{A}$ because the marginal unit of collateral is then of use to nobody. More of the asset pushes its own price down, and it is that price the externality tracks. The asset supply therefore matters only inside $(\underline{A}, \overline{A})$: the band in which the asset adds liquidity is exactly the band in which it distorts prices.

Separating two calculations is what makes the word *externality* the right one. Hold the equilibrium balance and secondary-market price fixed and raise A: this is the calculation an individual buyer performs, since she takes prices as given. Then κ is fixed, A enters profit per customer only through total liquidity on region I, and that profit can only rise – on region I because there is more to spend, and on the stretch that leaves region I as $\breve{\rho}$ falls, because the buyer there was constrained and is no longer. So F rises pointwise and the distribution of prices falls in the sense of first-order stochastic dominance: to her, more of the asset is an unambiguous gain. In general equilibrium the same experiment *raises* the mean posted price, because ψ_C falls and κ rises against region II. The gap between the two calculations is the price movement that no individual buyer takes into account.

One contrast closes the argument. In the scarce case the asset commands its largest possible liquidity premium: with $\psi_C = d$ it is a perfect substitute for money, and (2.27) together with (2.26) gives $\psi = \beta d(1+i)$ exactly, so the premium equals the nominal rate. In the plentiful case region I is empty, the integral in (2.27) is taken over an empty set, and $\psi = \beta d$: the premium is zero. The premium is therefore largest exactly where the externality is absent, and vanishes exactly where the externality is largest. A liquidity premium measures whether the asset substitutes for money, not how much liquidity it provides, and it is no guide at all to what the asset market does to prices.

3.2 Monetary policy and welfare implications

Section 3.1 held inflation fixed and moved the asset supply. Both cutoffs of Proposition 1 depend on μ, however, so the exercise can be run the other way: fix the asset supply and raise inflation, and the same economy passes through the same three regimes. Monetary policy selects the regime. What we ask of it here is not what it does to prices, which Section 3.1 has settled, but what it does to welfare, and we ask it in the same way: our economy against the benchmark of Head et al. (2012), at a common rate of inflation.

The criterion is the surplus generated in the decentralized market,

$$\mathcal{W} \;\equiv\; \int \varrho(\rho)\Big[u\big(q(\rho)\big) - c\,q(\rho)\Big]\,dF(\rho), \tag{3.4}$$

where ϱ is the measure of buyers served at ρ, from (2.16). So (3.4) adds the match surplus $u(q) - cq$ over every meeting that takes place, and the buyers who sample no firm contribute nothing. Lifetime welfare is $(1-\beta)^{-1}\big[\mathcal{W} + U(x^*) - x^*\big]$; quasi-linearity fixes x^* irrespective of policy, the lump-sum transfer nets against the inflation tax in the aggregate, and the asset is

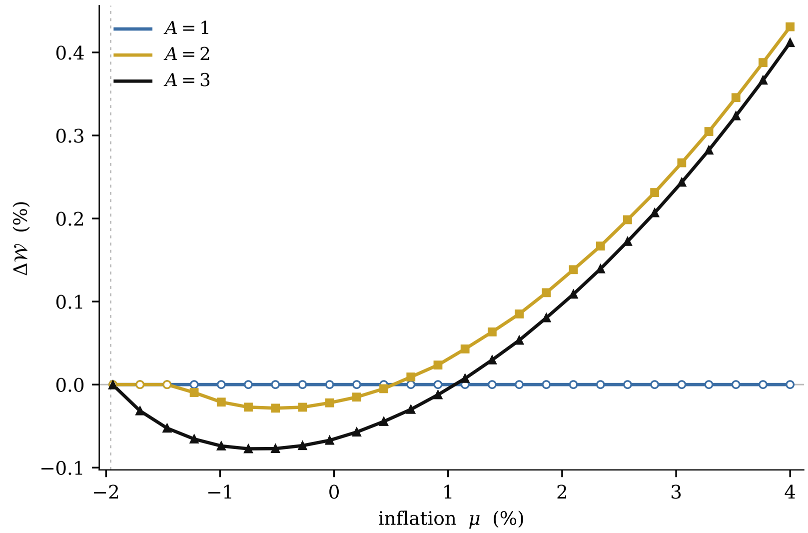


Figure 8: Welfare against the benchmark, along inflation.

in fixed supply, so the centralized term is a constant and lifetime welfare is ordered exactly as $\mathcal{W}$ is. Any profit earned by whoever issues the asset is likewise absent: issuers live only in the centralized market and their preferences are linear there, so such profit is a transfer between agents whose marginal utility of the general good is unity. Welfare comparisons are comparisons of $\mathcal{W}$ alone. We report $\Delta\mathcal{W}$, the percentage gap in (3.4) against the benchmark at the same rate of inflation.

Figure 8 draws it. The three curves are the three asset supplies of Figure 7, so the comparison is the one already made, now run along inflation rather than along the asset supply; the zero line is the benchmark, and below it the asset market lowers welfare. The markers give the regime at each point, and they move: a curve drawn at a fixed asset supply changes regime as inflation rises, because the cutoffs come down to meet it.

The scarce curve is a control carried inside the figure. It lies on the zero line at every rate of inflation, and it lies on it exactly. This is the top row of Figure 7 read as a function of policy: the asset trades at $\psi_C = d$, so the four terms of (2.28) collapse to one, the asset displaces money one for one and adds nothing, and the two economies coincide. The asset market is open at every rate of inflation and at every rate of inflation it does nothing. No monetary policy can separate the two economies while the asset is a perfect substitute for money.

The intermediate curve leaves the zero line, and it leaves it at a rate of inflation rather than at an asset supply. Inflation shrinks real balances; smaller balances mean less cash brought to the asset market by lenders; so $\underline{A}$ falls, and a supply that was scarce becomes intermediate without anybody changing it. From that point everything in Section 3.1 applies: ψ_C falls below d, the wedge opens, purchasing power moves from the middle of the price distribution to the bottom, the two distributions cross, and the buyer who pays out of her own money is quoted more than she would have been. The surplus falls with her.

The plentiful curve is the same thing, larger. Region I is empty at every rate of inflation here, so the asset supply has already left the equilibrium and the curve lies below the other two throughout the range in which the externality is negative. It is larger for the reason Section 3.1 gave: what orders the size of the effect is not how much of the asset there is but how far its price has fallen below the dividend, and that gap is widest here. At the Friedman rule every curve returns to the benchmark, because the admissible interval for ψ_C closes onto d and the wedge is squeezed to zero.

The three curves together give the result of this section. *More of the asset switches the externality on at a lower rate of inflation, makes it deeper, and postpones the rate at which it turns into a gain.* At low inflation the surplus falls, and it falls by more the more of the asset there is. The ordering is the one Section 3.1 established along the asset supply, and it stops where that one stopped: beyond $\overline{A}$ a larger supply traces the same curve.

The mechanism is that of Kam et al. (2024), and the difference is what the buyer is holding. There the instrument is bank credit, and firms with market power extract the liquidity that credit supplies, at the expense of the buyers who do not borrow. Here the instrument is an asset, and firms extract the liquidity that the secondary market supplies, at the expense of the buyers who do not sell. In both the loser is the buyer who does not use the instrument at all. What the asset adds is a second margin. Credit is created on demand at a rate; an asset is in fixed supply and carries a price of its own, and that price is what the externality tracks. So the asset supply becomes a policy dimension alongside inflation, and it is a bounded one: the externality grows with the asset only until the asset stops being scarce, and then stops growing altogether.

4 Endogenous asset supply and the pecuniary externality

The asset supply has been a parameter. We now let it be chosen, and ask whether the choice is the right one. The externality of Section 3.2 is a price effect that no buyer takes into account; an issuer does not take it into account either, and the question is how far that carries him from the social optimum. The formulation of the issuer's problem follows Geromichalos et al. (2023), who study the strategic supply of liquid assets in a related environment; what is new here is that the goods market carries a distribution of prices, so that the issuer's decision moves not only the level of liquidity but the dispersion of the prices at which it is spent.

Everything in this section is an object evaluated *along the equilibrium correspondence*. For a given A, Definition 1 determines $\big(z(A), \psi_C(A)\big)$ and hence, through (2.22), the distribution F_A and the demand q_A of Lemma 1. Two objects then do all the work. The first is the surplus $\mathcal{W}(A)$, which is (3.4) evaluated at F_A and q_A. The second is what the asset earns whoever issues it. A unit is a one-period claim: it is sold in the centralized market at $\psi(A)$, which is (2.27) read as a function of A, and obliges its issuer to deliver d in the next centralized market, a liability worth βd today. The difference is the liquidity premium, and it is the issuer's profit per unit,

$$L(A) \;\equiv\; \psi(A) - \beta d \;\geq\; 0. \tag{4.1}$$

Three parties might choose the asset supply, and each solves a different problem. A monopolist issuer knows that another unit depresses the price at which he sells every unit he has issued, so the premium enters his problem as a function and he solves

$$A^m \;\in\; \arg\max_{A \geq 0}\; \Pi(A), \qquad \Pi(A) \equiv L(A)\, A. \tag{4.2}$$

Competitive issuers take the asset price as given, so for each of them the premium is a number rather than a function; each earns L on the marginal unit and entry continues while that is positive, so the supply they deliver satisfies the free-entry condition

$$L\big(A^c\big) \;=\; 0. \tag{4.3}$$

A planner chooses the asset supply and nothing else: she may not dictate prices, may not choose z, and may not alter the posting game. She selects A, the private sector responds with the equilibrium of Definition 1, and she solves

$$A^p \;\in\; \arg\max_{A \geq 0}\; \mathcal{W}(A). \tag{4.4}$$

That last restriction is what makes her the right benchmark for a pecuniary externality. She is granted no instrument the market lacks, so any gap between her choice and the decentralized ones is a failure to internalize a price rather than a failure of market structure to replicate a first best. In particular $\mathcal{W}$ remains bounded away from the first best at every A, since by Lemma 2 every buyer pays a markup.

Comparing the three shows where the wedge sits. The two decentralized problems involve only L and its slope — the price at which the asset sells and how that price responds — while the planner's involves $\mathcal{W}'$. An issuer internalizes the private value of the liquidity he creates, because that value is capitalized into ψ and he collects it. What he does not internalize is the effect of his issue on ψ_C, which by Section 3.1 moves κ and hence every buyer's real balances. Nothing forces L and $\mathcal{W}'$ to agree, and Section 3.2 has already shown that they do not: at any $A < \underline{A}$ we have $L = \beta d\, i > 0$ while $\mathcal{W}' = 0$.

Proposition 1 already determines the shape of L, and the shape is what settles both market structures. Below $\underline{A}$ the asset market is idle, $\psi_C = d$, and by the argument in Section 3.1 the asset price is $\beta d(1 + i)$; the premium is therefore

$$L(A) = \beta d\, i \qquad \text{for all } A < \underline{A}(\mu), \tag{4.5}$$

constant in the asset supply and equal to the nominal rate per unit of fundamental value. Above $\overline{A}$ region I is empty, the integral in (2.27) is taken over an empty set, and $L = 0$. In between L falls from the one value to the other.

Proposition 2 (Decentralized supply)**.** Fix $\mu > \beta - 1$ and let issuers bear no cost beyond the dividend obligation.

(i) a monopolist issuer, who solves $\max_{A \geq 0} L(A)\, A$, chooses $A^m = \underline{A}(\mu)$ and earns $\beta d\, i\, \underline{A}(\mu)$;

(ii) competitive issuers, who enter while $L > 0$, issue until the premium is exhausted, so that $A^c = \overline{A}(\mu)$;

(iii) a planner who solves $\max_{A \geq 0} \mathcal{W}(A)$ subject to the equilibrium of Definition 1 may take her choice in $[\underline{A}(\mu), \overline{A}(\mu)]$.

The proof is in the appendix. Putting the three solutions on one line is the comparison this section exists to make:

$$A^m \;=\; \underline{A}(\mu) \;\leq\; A^p \;\leq\; \overline{A}(\mu) \;=\; A^c, \qquad \mathcal{W}\big(A^m\big) \;=\; \mathcal{W}_0 \;\leq\; \mathcal{W}\big(A^p\big), \tag{4.6}$$

where $\mathcal{W}_0$ is the surplus of the benchmark with the asset market closed. The two decentralized outcomes are not merely different from the optimum; they are the two *endpoints* of the interval that contains it. A monopolist stops exactly where the asset market would begin to operate, and competitive issuers stop exactly where it ceases to matter. The band in which the asset supply does anything at all is bracketed by the two market structures, and the planner is somewhere

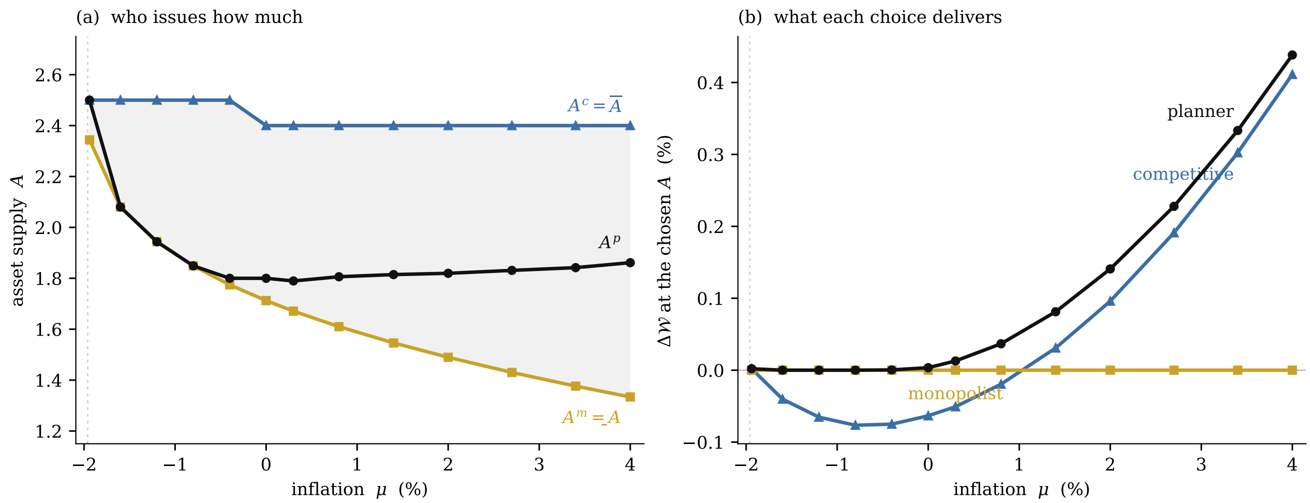


Figure 9: The three asset supplies and what they deliver, against inflation.

inside. The second half of (4.6) says that the monopolist's choice delivers the benchmark exactly, and that the planner can never do worse, since A^m is feasible for her.

What the monopolist is protecting is worth naming. By (4.5) his margin is at its maximum, $\beta d\, i$, on every unit he sells so long as $\psi_C = d$, and it collapses the moment one more unit pushes ψ_C below d. He therefore issues as much as he can *without opening the asset market*: he chooses to keep the asset a perfect substitute for money. Competitive issuers, each taking ψ as given, run the premium to zero and push the economy into regime 3, where by Section 3.1 the deviation from the benchmark is largest.

Panel (a) of Figure 9 draws the three choices across inflation: the monopolist's is marked by squares, the competitive outcome by triangles, and the constrained optimum by circles, with the vertical dotted line at the Friedman rule throughout. It is Proposition 2 in one picture, since the shaded band between $\underline{A}$ and $\overline{A}$ is simultaneously the region in which the asset supply matters and the interval spanned by the two market structures, with the planner inside it. The planner's choice is computed on a grid in A and refined by a local quadratic fit, and is reported inside that band, which Proposition 2(iii) permits; near the Friedman rule the surplus is flat across the whole band, so its location there carries no information and only its value, which is zero, does.

Panel (b) shows what those choices deliver, measured as in Figure 8 against the benchmark with the asset market closed, and the monopolist's curve is the flat one at zero. That is not a numerical accident but a consequence of $A^m = \underline{A}$: at his choice the asset market is idle, so by Proposition 1(i) the decentralized market is exactly the benchmark. Under a monopolist issuer the asset exists, is held, and is traded, and yet the distribution of prices is the one that would prevail with no asset market at all — at every rate of inflation.

Why the monopolist should be the one who stops is worth stating, because it inverts the usual reading. His objective $L(A)A$ contains $L'(A) < 0$: he knows that another unit depresses the secondary-market price and with it the premium on every unit he has already sold. He therefore *does* internalize the price effect — it is his revenue. A competitive issuer does not, because for him ψ is a number rather than a function, and L' is not an object in his problem at all. The externality of Section 3.1 is precisely this price effect, so the market structure that is ordinarily the distortion is here the only arrangement in which anybody accounts for it. Monopoly does not internalize it for the right reason, and it internalizes the wrong magnitude, but the sign is correct.

The consequence is that the two structures deliver the two extremes of Section 3.1. A monopolist lands on $\underline{A}$, hence regime 1, where the deviation from the benchmark is exactly zero. Competitive issuers land on $\overline{A}$, hence regime 3, where by Section 3.1 it is largest. *The size of the pecuniary externality is set by the market structure for issuance*, and free entry maximizes it.

Which of the two is the less costly depends on the sign of the externality, and the sign is monetary policy's. At low inflation the asset market damages the decentralized market, so running the premium to zero is the worst thing to do and the monopolist's restriction costs nothing; at high inflation the ranking is reversed. The two losses cross, and they cross at the inflation rate at which the externality changes sign in Figure 8. The coincidence is not one: restricting the supply of a liquid asset is socially valuable exactly when the asset market lowers the surplus, and both are decided by the same feature of the equilibrium, namely whether the crowding out of real balances outweighs the liquidity the asset provides. Monetary policy settles that, and in settling it also settles which market structure for issuance is the less costly.

Two forces widen the gap as inflation rises, and they push the same way. The monopolist's choice is $\underline{A}(\mu)$, which falls with inflation, because inflation lowers the cash lenders bring and the asset market opens at a smaller collateral stock. The planner's choice does not fall with it. At low rates she too leaves the asset market idle, since every larger supply lowers the surplus; past the rate at which the externality changes sign she moves inside the band and stays there. *Private supply therefore contracts exactly as the social value of the asset is rising*, and the monopoly loss grows with inflation from nothing to its largest value at the top of the range.

We record two limitations. First, the issuer here bears no cost beyond the dividend, which is what makes both decentralized outcomes land on the cutoffs; a convex issuance cost would move A^c inside the band, towards the optimum, without changing the monopolist's corner. Second, $A^m = \underline{A}$ leaves the asset market idle, so the monopolist's equilibrium is the knife-edge case in which lenders are indifferent and the participation constraint (2.15) holds with equality. Endogenizing the measure of lenders who participate would pin down that margin rather than leaving it on a boundary, and we leave it to future work.

5 Conclusion

We have asked what a competitive over-the-counter market does to a goods market in which prices are posted and dispersed. A buyer who expects to be able to sell assets for cash carries less cash out of the centralized market, and a buyer who then draws an expensive quote is left with the smaller balance and nothing to supplement it. Firms with market power see both kinds of buyer and post accordingly. The question is what that does to the distribution of prices, and to whom.

The answer is that the liquidity service of the asset market is not neutral. While the asset trades at its dividend it displaces money one for one and changes nothing at all. Once it trades below the dividend the distribution of prices shifts, and it shifts against the buyer who never enters the asset market. At low rates of inflation this lowers welfare, and it lowers it by more the more plentiful the asset, until the asset ceases to be scarce and the effect stops growing. When the supply is chosen rather than given, a monopolist issuer internalizes the price effect and eliminates the externality, while competitive issuers put more of the asset into circulation and maximize it.

Two restrictions point to the same next step. Our issuer bears no cost beyond the dividend, which is what places the decentralized outcomes exactly on the cutoffs, and the monopolist leaves the asset market idle, so his equilibrium sits on a knife-edge at which lenders are indifferent about participating. Replacing the competitive asset market with the search-and-bargaining frictions of Duffie et al. (2005) would resolve both, and would add a second source of dispersion: one friction in the goods market, which disperses the price a buyer pays, and another in the asset market, which disperses the price at which she can raise cash. Whether such frictions amplify the externality identified here or offset it is a question the competitive benchmark cannot answer.

Appendix

Proof of Lemma 1

The objective in (2.11) is $u(q) - \beta\rho q - g\,\chi$ up to a constant, strictly concave in q and linear in χ, and the constraint set is convex, so the Kuhn–Tucker conditions are necessary and sufficient. Attach λ_1 to $\rho q \leq z + \chi$, λ_2 to $\chi \geq 0$ and λ_3 to $\chi \leq \psi_C a$. The first-order conditions are

$$u'(q) = \big(\beta + \lambda_1\big)\rho, \qquad\qquad \lambda_3 - \lambda_2 = \lambda_1 - g, \tag{.1}$$

with the usual complementary slackness conditions. The second condition determines which constraint binds according to the position of λ_1 relative to the band $[0, g]$, and yields four cases.

1. $\lambda_1 = 0$. Then $\lambda_3 - \lambda_2 = -g < 0$, so $\lambda_2 = g > 0$ and $\chi = 0$. The first condition gives $u'(q) = \beta\rho$, hence $q = (\beta\rho)^{-1/\sigma}$. This is consistent with $\rho q \leq z$ if and only if $\rho \geq \hat{\rho}$, where $\hat{\rho}$ solves $\rho(\beta\rho)^{-1/\sigma} = z$, that is $\hat{\rho} = \beta^{1/(\sigma-1)} z^{\sigma/(\sigma-1)}$.

2. $0 < \lambda_1 < g$. Then $\lambda_2 = g - \lambda_1 > 0$ and $\lambda_3 = 0$, so $\chi = 0$ and $\rho q = z$. The first condition gives $u'(z/\rho) = (\beta + \lambda_1)\rho$, and $\lambda_1 \in (0, g)$ requires $\beta\rho < u'(z/\rho) < \kappa\rho$, which holds if and only if $\tilde{\rho} \leq \rho < \hat{\rho}$ with $\tilde{\rho} = \kappa^{1/(\sigma-1)} z^{\sigma/(\sigma-1)}$.

3. $\lambda_1 = g$. Then $\lambda_2 = \lambda_3 = 0$ and χ is interior. The first condition gives $u'(q) = \kappa\rho$, hence $q = (\kappa\rho)^{-1/\sigma}$ and $\chi = \rho q - z$. This is admissible, $0 \leq \chi \leq \psi_C a$, if and only if $\breve{\rho} \leq \rho < \tilde{\rho}$ with $\breve{\rho} = \kappa^{1/(\sigma-1)} X^{\sigma/(\sigma-1)}$, obtained by setting $\rho q = X$.

4. $\lambda_1 > g$. Then $\lambda_3 = \lambda_1 - g > 0$, so $\chi = \psi_C a$ and $\rho q = X$, giving $q = X/\rho$. This requires $u'(X/\rho) > \kappa\rho$, that is $\rho < \breve{\rho}$.

Ordering. Since $\sigma \in (0, 1)$, both $1/(\sigma - 1)$ and $\sigma/(\sigma - 1)$ are negative. Since $X > z$, $X^{\sigma/(\sigma-1)} < z^{\sigma/(\sigma-1)}$ and therefore $\breve{\rho} < \tilde{\rho}$. Since $g > 0$ is equivalent to $\kappa > \beta$, we have $\kappa^{1/(\sigma-1)} < \beta^{1/(\sigma-1)}$ and therefore $\tilde{\rho} < \hat{\rho}$. Conversely $\tilde{\rho} < \hat{\rho}$ implies $\kappa > \beta$, and $\kappa > \beta$ is equivalent to $\psi_C < d$, since $\kappa = \beta d/\psi_C$. This proves the last statement.
Continuity. At $\breve{\rho}$ the two adjacent branches agree: using $\kappa\,\kappa^{1/(\sigma-1)} = \kappa^{\sigma/(\sigma-1)}$,

$$(\kappa\breve{\rho})^{-1/\sigma} = \Big(\kappa^{\frac{\sigma}{\sigma-1}} X^{\frac{\sigma}{\sigma-1}}\Big)^{-1/\sigma} = (\kappa X)^{-\frac{1}{\sigma-1}}, \qquad \frac{X}{\breve{\rho}} = \kappa^{-\frac{1}{\sigma-1}} X^{\frac{(\sigma-1)-\sigma}{\sigma-1}} = (\kappa X)^{-\frac{1}{\sigma-1}}.$$

The same computation with X replaced by z gives continuity at $\tilde{\rho}$, and with κ replaced by β gives continuity at $\hat{\rho}$. This proves part (i).

Part (ii). Let $g = 0$, so that $\kappa = \beta$. The four cases of the Kuhn–Tucker conditions above collapse to two, because the band $[0, g]$ within which λ_1 was compared has shrunk to the single point $\{0\}$. Cases 2 and 3, which required $0 < \lambda_1 < g$ and $\lambda_1 = g > 0$, are empty. Case 1 gives $\lambda_1 = 0$, hence $u'(q) = \beta\rho$ and $q = (\beta\rho)^{-1/\sigma}$; case 4 gives $\lambda_1 > 0$, hence $\chi = \psi_C a$ and $q = X/\rho$.

The same conclusion can be read off the cutoffs. Setting $\kappa = \beta$ in (2.13) gives $\tilde{\rho} = \beta^{1/(\sigma-1)} z^{\sigma/(\sigma-1)} = \hat{\rho}$, so the interval $[\tilde{\rho}, \hat{\rho})$ on which region III was defined is empty, and it gives $\breve{\rho} = \beta^{1/(\sigma-1)} X^{\sigma/(\sigma-1)}$. On $[\breve{\rho}, \tilde{\rho})$ the region-II branch is $(\kappa\rho)^{-1/\sigma} = (\beta\rho)^{-1/\sigma}$, which is the region-IV branch, so the two join into a single expression valid on $[\breve{\rho}, \infty)$. Demand is therefore X/ρ below $\breve{\rho}$ and $(\beta\rho)^{-1/\sigma}$ above it, which is the demand of Head et al. (2012) with X in the place of real balances.

Financing does not collapse with it. From the budget constraint $\chi = \max\{0, \rho q - z\}$, subject to the cap $\psi_C a$. Below $\breve{\rho}$ the cap binds and $\chi = \psi_C a$. Above it $\rho q = \rho(\beta\rho)^{-1/\sigma} = \beta^{-1/\sigma}\rho^{1-1/\sigma}$, which is decreasing in ρ because $\sigma < 1$, and equals z exactly at $\hat{\rho}$. Hence $\chi = \rho(\beta\rho)^{-1/\sigma} - z$ is positive on $[\breve{\rho}, \hat{\rho})$ and zero on $[\hat{\rho}, \infty)$. Continuity at $\breve{\rho}$ follows as before, since $\breve{\rho} q = X$ there and $X - z = \psi_C a$; at $\hat{\rho}$ both expressions equal zero. $\square$

Proof of Lemma 2

Step 1: the shape of R and the candidate set.

Substituting Lemma 1 into $R(\rho) = q(\rho)(\rho - c)$ gives four branches. On $[c, \breve{\rho})$ and on $[\tilde{\rho}, \hat{\rho})$ the buyer spends a fixed sum, so $R = W(1 - c/\rho)$ with $W = X$ and $W = z$ respectively, and

$$R'(\rho) = \frac{Wc}{\rho^2} > 0 :$$

both branches are strictly increasing and have no interior stationary point. On $[\breve{\rho}, \tilde{\rho})$ and on $[\hat{\rho}, \infty)$ she equates marginal utility to a constant times the price, so $R = \theta^{-1/\sigma}\rho^{-1/\sigma}(\rho - c)$ with $\theta = \kappa$ and $\theta = \beta$ respectively, and

$$R'(\rho) = \frac{\theta^{-1/\sigma}}{\sigma}\, \rho^{-\frac{1}{\sigma}-1}\big[c - (1-\sigma)\rho\big],$$

which is positive for $\rho < \rho^*$ and negative for $\rho > \rho^*$, *whatever* θ: the elasticity of demand is $1/\sigma$ on both branches, so θ scales R without moving its peak. Since $\sigma < 1$ we have $R \sim \theta^{-1/\sigma}\rho^{1-1/\sigma} \to 0$ as $\rho \to \infty$, so the maximum is attained.

R is continuous, by the continuity of q established in the proof of Lemma 1. A local maximum can therefore occur only where R rises from the left and falls to the right, which leaves three possibilities: ρ^*, when it lies in one of the two single-peaked branches; $\breve{\rho}$, when R rises into it and falls out, which requires $\breve{\rho} > \rho^*$; and $\hat{\rho}$, on the same condition. At $\tilde{\rho}$ the branch to the right is increasing, so $\tilde{\rho}$ is never a local maximum.

The two humps of R are $[c, \tilde{\rho})$ and $[\tilde{\rho}, \infty)$. On the first, R increases to $\breve{\rho}$ and is then single-peaked at ρ^*, so its supremum is attained at ρ^* if $\breve{\rho} \le \rho^* < \tilde{\rho}$, at $\breve{\rho}$ if $\rho^* < \breve{\rho}$, and approached at $\tilde{\rho}$ if $\rho^* \ge \tilde{\rho}$; these three are $\rho_\kappa = \min\{\max\{\rho^*, \breve{\rho}\}, \tilde{\rho}\}$. On the second, R increases to $\hat{\rho}$ and is then single-peaked at ρ^*, so its maximizer is $\rho_\beta = \max\{\rho^*, \hat{\rho}\}$. This is (2.19). Since the first paragraph already restricts the local maxima to $\{\breve{\rho}, \rho^*, \hat{\rho}\}$, and each of ρ_κ, ρ_β is one of those three or is the dominated $\tilde{\rho}$, taking the maximum of R over the three prices directly is equivalent; that is (2.18). It remains to check that comparing the two humps delivers the stated ρ^m in each configuration of ρ^* relative to the cutoffs.

1. $\rho^* \ge \hat{\rho}$. Every branch below $\hat{\rho}$ is increasing, and the last branch increases up to ρ^* and falls thereafter, so R is increasing on $[c, \rho^*]$. Here $\rho_\kappa = \tilde{\rho}$ and $\rho_\beta = \rho^*$, and $R(\tilde{\rho}) < R(\rho^*)$ because R rises between them, so $\rho^m = \rho^*$.

2. $\tilde{\rho} \le \rho^* < \hat{\rho}$. Here ρ^* falls in the third branch, which has no stationary point, so R is increasing on $[c, \hat{\rho}]$ and decreasing thereafter. Here $\rho_\kappa = \tilde{\rho}$ and $\rho_\beta = \hat{\rho}$, and $R(\tilde{\rho}) < R(\hat{\rho})$, so $\rho^m = \hat{\rho}$.

3. $\breve{\rho} \le \rho^* < \tilde{\rho}$. Here $\rho_\kappa = \rho^*$ and $\rho_\beta = \hat{\rho}$, both are local maxima, and neither dominates: ρ^m is whichever earns the larger R.

4. $\rho^* < \breve{\rho}$. Here $\rho_\kappa = \breve{\rho}$ and $\rho_\beta = \hat{\rho}$, again both local maxima, and again the comparison decides.

Finally, the closed forms. Evaluating R on the relevant branch gives $R(\breve{\rho}) = (X/\breve{\rho})(\breve{\rho} - c) = X(1 - c/\breve{\rho})$ and $R(\hat{\rho}) = z(1 - c/\hat{\rho})$. At $\rho^* = c/(1-\sigma)$ we have $\rho^* - c = c\sigma/(1-\sigma)$ and $(\rho^*)^{-1/\sigma} = c^{-1/\sigma}(1-\sigma)^{1/\sigma}$, so

$$R(\rho^*) = \theta^{-1/\sigma}\, c^{-1/\sigma}(1-\sigma)^{1/\sigma} \cdot \frac{c\sigma}{1-\sigma} = \theta^{-1/\sigma}\,\sigma\, c^{\frac{\sigma-1}{\sigma}}(1-\sigma)^{\frac{1-\sigma}{\sigma}} = \Phi\, \theta^{-1/\sigma},$$

with $\theta = \kappa$ when ρ^* lies on region II and $\theta = \beta$ when it lies on region IV.

Step 2: the classification. Take the first equivalence. By (2.13), $\hat{\rho} = \beta^{1/(\sigma-1)} z^{\sigma/(\sigma-1)}$, so $\hat{\rho} \le \rho^*$ iff $\beta^{1/(\sigma-1)} z^{\sigma/(\sigma-1)} \le \rho^*$. Raising both sides to the power $(\sigma-1)/\sigma$, which is negative because $\sigma < 1$ and therefore reverses the inequality, gives $\beta^{1/\sigma} z \ge (\rho^*)^{(\sigma-1)/\sigma}$, that is $z \ge z_\beta$. The same computation applied to $\tilde{\rho}$ gives $\tilde{\rho} \le \rho^* \iff z \ge z_\kappa$, and applied to $\breve{\rho}$ gives $\breve{\rho} \le \rho^* \iff X \ge z_\kappa$, which since $X = z + \psi_C a$ is $a \ge (z_\kappa - z)/\psi_C$. Since $\theta \mapsto \theta^{-1/\sigma}$ is decreasing and $\kappa \ge \beta$ whenever $\psi_C \le d$, we have $z_\kappa \le z_\beta$, and because $X > z$ the four listed conditions are exhaustive and mutually exclusive.

The value of ρ^m in each of the four regions is read off the four configurations of Step 1. In region IV, $z \geq z_\beta$ gives $\hat{\rho} \leq \rho^*$, which is configuration 1, and $\rho^m = \rho^*$. In region III, $z_\kappa \leq z < z_\beta$ gives $\tilde{\rho} \leq \rho^* < \hat{\rho}$, configuration 2, and $\rho^m = \hat{\rho}$. Regions II and I are configurations 3 and 4, in which two local maxima survive.

Step 3: the case $g = 0$. Then $\kappa = \beta$, so $z_\kappa = z_\beta$ and the interval defining region III is empty. By Lemma 1(ii) the branch on $[\breve{\rho}, \tilde{\rho})$ and the branch on $[\hat{\rho}, \infty)$ carry the same expression and $\tilde{\rho} = \hat{\rho}$, so the two single-peaked branches are one branch on $[\breve{\rho}, \infty)$, peaking at ρ^*. The two humps of R are then one hump, and ρ_κ and ρ_β both reduce to the maximizer of that single hump, $\max\{\rho^*, \breve{\rho}\}$, which is therefore ρ^m. $\square$

Proof of Proposition 1

Throughout, $\mathcal{R}(z, \psi_C)$ is the liquidity premium (2.28), so that (2.26) reads $i = \mathcal{R}(z, \psi_C)$ with $i \equiv (1+\mu)/\beta - 1$. Write $\mathcal{Z}(\psi_C)$ for its solution in z and

$$E(\psi_C) \;\equiv\; \underbrace{\int \big[\alpha_1 + 2\alpha_2(1-F)\big]\chi(\rho)\,dF(\rho)}_{\text{cash borrowers raise}} \;-\; \underbrace{\alpha_0\,\mathcal{Z}(\psi_C)}_{\text{cash lenders bring}}$$

for the excess demand for cash, $\chi(\rho)$ being the asset the buyer sells, as in Lemma 1. Every object in Definition 1 is continuous in (z, ψ_C): the cutoffs of Lemma 1 and the candidates of Lemma 2 are continuous, F depends on ρ^m only through $R(\rho^m) = \max_\rho R(\rho)$, which is continuous even where the argmax jumps between two tied local maxima, and $\bar{R}$ is a running maximum of a continuous function. Hence $\mathcal{R}$ and E are continuous.

Step 1. Fix ψ_C. As $z \to 0$ every buyer is constrained at every price in the support and $u'(q(\rho)) \to \infty$, so the region I and region III terms of (2.28) diverge. As $z \to \infty$ the cutoffs $\breve{\rho}, \tilde{\rho}, \hat{\rho}$ fall below $\underline{\rho}$, regions I–III carry no mass, and $\mathcal{R}$ falls to the no-quote term $\alpha_0 g/\beta$. Since $\mathcal{R}$ is continuous, a solution to $i = \mathcal{R}(z, \psi_C)$ exists if and only if the nominal rate exceeds the return to lending idle cash,

$$i > \frac{\alpha_0\, g}{\beta} \qquad\Longleftrightarrow\qquad \psi_C > \psi_C^{\min} \equiv \frac{\alpha_0\, d}{\alpha_0 + i}. \tag{.2}$$

Below $\psi_C^{\min}$ lending beats the inflation tax and money demand is unbounded. By hypothesis $\mathcal{R}$ is decreasing in z, so that solution is unique; $\mathcal{Z}$ is therefore a function and, being the solution of a continuous strictly monotone equation, is continuous. This is the only place the hypothesis is used before the last step, and it is what makes E a function of ψ_C alone.

Step 2. As $\psi_C \downarrow \psi_C^{\min}$ we have $\mathcal{R}(z, \psi_C) \downarrow \alpha_0 g/\beta \uparrow i$, so $\mathcal{Z}(\psi_C) \to \infty$. Real balances then exceed what any buyer wishes to spend, $\tilde{\rho}$ falls below $\underline{\rho}$, no buyer sells assets and the demand for cash is zero, while supply $\alpha_0 \mathcal{Z}$ diverges; hence $E(\psi_C) \to -\infty$. At the other end $E(d)$ is finite.

Step 3. If $E(d) > 0$ then, E being continuous and negative near $\psi_C^{\min}$, the intermediate value theorem gives a root $\psi_C^* \in (\psi_C^{\min}, d)$; the complementary-slackness conditions (2.15) hold with $\psi_C^* < d$ and $S = \alpha_0 z$. If $E(d) \leq 0$ then $\psi_C = d$ with $S = \int [\alpha_1 + 2\alpha_2(1-F)]\chi \, dF \leq \alpha_0 z$ satisfies (2.15), the second factor being slack and the first zero. Either way an equilibrium exists.

Uniqueness. $\mathcal{R}$ is decreasing in z by hypothesis and is also decreasing in ψ_C, since raising ψ_C lowers κ and with it the no-quote and region II terms of (2.28); the implicit function $\mathcal{Z}$ is therefore decreasing in ψ_C. If in addition the demand for cash is increasing in ψ_C, then E is the sum of an increasing function and $-\alpha_0 \mathcal{Z}$, itself increasing, so E is strictly increasing and has at most one root. □

Proof of Lemma 4

Write $M(z) \equiv R(\rho^m; z)$, so that $\Pi^* = \alpha_1 M(z)$ and (2.22) reads $F(\rho; z) = 1 - \frac{\alpha_1}{2\alpha_2}\left[\Lambda(\rho; z)^{-1} - 1\right]$ with $\Lambda = \bar{R}/M$. Every claim about F is the reverse claim about Λ. The difficulty the proof has to meet is that the cutoffs (2.13) and the maximizer (2.18) all move with z: a price at which the buyer with z_1 is unconstrained may be one at which the buyer with $z_0 < z_1$ is not, and the two economies may compare their profits at different maximizers. Steps 3 and 4 are arranged so that neither has to be tracked.

Step 1: profit per customer above $\tilde{\rho}$ is a minimum. Write $G(\rho) \equiv (\beta\rho)^{-1/\sigma}(\rho - c)$ for the profit a firm earns from a buyer who is not liquidity constrained. By Lemma 1 a buyer facing $\rho \geq \tilde{\rho}$ raises no cash and buys $q = z/\rho$ if $\rho < \hat{\rho}$ and $q = (\beta\rho)^{-1/\sigma}$ if $\rho \geq \hat{\rho}$, so R is $z(1 - c/\rho)$ on the first range and $G(\rho)$ on the second. Dividing both expressions by $(\rho - c)/\rho > 0$ turns the comparison between them into z against $\beta^{-1/\sigma}\rho^{(\sigma-1)/\sigma}$, and since $(\sigma - 1)/\sigma < 0$ the second is decreasing in ρ and equals z exactly at $\rho = \hat{\rho}$ by (2.13). The branch in force is therefore always the smaller of the two, and

$$R(\rho; z) = \min\left\{z\,(1 - c/\rho),\ G(\rho)\right\}, \qquad \rho \geq \tilde{\rho}. \tag{.3}$$

Only the first argument contains the real balance, and it contains it linearly.

Step 2: the maximizer lies in the second range. Suppose some $\rho \geq \tilde{\rho}$ is posted, so that $\rho^m \geq \rho \geq \tilde{\rho}$. In (2.18) the first candidate satisfies $\rho_\kappa \leq \tilde{\rho}$ by construction and the second satisfies

$\rho_\beta = \max\{\rho^*, \hat{\rho}\} \geq \hat{\rho} > \tilde{\rho}$, so $\rho^m = \rho_\beta = \max\{\rho^*, \hat{\rho}\}$ unless $\rho^m = \rho_\kappa = \tilde{\rho}$, in which case the range in question is the single point $\tilde{\rho}$ and there is nothing to prove. Hence $M(z) = R(\rho^m; z)$ is itself given by (.3).

Step 3: the elasticity of R in z is non-increasing in the price. Fix z and let $\rho \geq \tilde{\rho}$ with $\rho > c$, so that $R(\rho; z) > 0$ and its logarithm is defined. By Lemma 3 the support of F lies strictly above c, so this excludes no price that is ever posted. By (.3),

$$\frac{\partial \log R(\rho; z)}{\partial \log z} = \begin{cases} 1, & \rho < \hat{\rho}(z), \\ 0, & \rho > \hat{\rho}(z), \end{cases}$$

because the binding argument of the minimum is the first below $\hat{\rho}$ and the second above it. A constrained buyer converts an extra unit of money into revenue one for one; an unconstrained buyer converts none of it. Since $\hat{\rho}(z)$ is a single cutoff, this elasticity is non-increasing in ρ. Consequently, for any $\rho \leq \rho'$ in that range, the difference $\log R(\rho; z) - \log R(\rho'; z)$ has non-negative derivative in z, so

$$z \longmapsto \frac{R(\rho; z)}{R(\rho'; z)} \qquad \text{is non-decreasing, for every } \max\{\tilde{\rho}, c\} < \rho \leq \rho'. \tag{.4}$$

This is the whole of the argument: profit per customer is log-submodular in the price and the real balance, because money helps a firm only where the buyer it faces is constrained, and she is constrained only at the lower prices.

Step 4: the comparison. Let $z_0 < z_1$, let $\rho \geq \tilde{\rho}(z_0)$ lie in the support of $F(\cdot; z_0)$, so that $\rho > c$, and write ρ_j^m for the maximizer in economy j; both maximizers exceed c, since $\rho^m \geq \rho^* > c$ when $\rho^m = \rho^*$ and $\rho^m = \hat{\rho} > \rho^* > c$ otherwise. Since $\tilde{\rho}$ is decreasing in z, $\rho \geq \tilde{\rho}(z_1)$ as well, so (.4) applies in both. Because $\rho \leq \rho_1^m$, taking $\rho' = \rho_1^m$ in (.4) gives

$$\frac{R(\rho; z_0)}{R(\rho_1^m; z_0)} \;\leq\; \frac{R(\rho; z_1)}{R(\rho_1^m; z_1)} = \frac{R(\rho; z_1)}{M(z_1)} \;\leq\; \Lambda(\rho; z_1),$$

the last step because $\bar{R} \geq R$. Since ρ_0^m maximizes $R(\cdot; z_0)$ we have $R(\rho_1^m; z_0) \leq M(z_0)$, and since ρ is in the support of $F(\cdot; z_0)$ we have $\bar{R}(\rho; z_0) = R(\rho; z_0)$. Therefore

$$\Lambda(\rho; z_0) = \frac{R(\rho; z_0)}{M(z_0)} \;\leq\; \frac{R(\rho; z_0)}{R(\rho_1^m; z_0)} \;\leq\; \Lambda(\rho; z_1),$$

and $F(\rho; z_0) \leq F(\rho; z_1)$. Note that nothing here required the two economies to be in the same case of Lemma 2, nor ρ to lie in the same region of Lemma 1 in both.

Step 5: the closed forms. By Step 2, either $\rho^* \geq \hat{\rho}$ and $\rho^m = \rho^*$, which by Lemma 2 makes $M = \Phi\beta^{-1/\sigma}$, or $\rho^m = \hat{\rho}$ and $M = z(1 - c/\hat{\rho})$. Dividing (.3) by M in each gives the three lines of (2.24). In the second line the factor $\beta^{-1/\sigma}$ in G cancels against the one in M, leaving $\Lambda = \Phi^{-1}\rho^{-1/\sigma}(\rho - c)$, which contains neither z nor ψ_C nor a nor β; inverting F delivers (2.25). In the third line the real balance cancels between numerator and denominator and survives only through $\hat{\rho}$. The monotonicity asserted in the lemma is immediate from these expressions: the first line is linear and increasing in z, the second is constant, and in the third $\hat{\rho}$ is decreasing in z by (2.13), so a smaller z raises $1 - c/\hat{\rho}$ and lowers Λ. This proves part (i).

Part (ii). Let $z_\beta \leq z_0 < z_1$; since region IV of Lemma 2 is a lower bound on z, it holds at both, so $M(z_0) = M(z_1) = \Phi\beta^{-1/\sigma}$ and it is enough that $\bar{R}(\rho; \cdot)$ be non-decreasing at every ρ, not merely above $\tilde{\rho}$. Fix ρ. By Lemma 1, $R = X(1 - c/\rho)$ on region I, $R = (\kappa\rho)^{-1/\sigma}(\rho - c)$ on II, $R = z(1 - c/\rho)$ on III and $R = G(\rho)$ on IV. The first is increasing in z through $X = z + \psi_C a$ and the third is increasing in z; the second and fourth do not contain z. As z varies the cutoffs move, so ρ may pass from one region to another, but q and hence R are continuous in z across each cutoff, being continuous in z at fixed ρ on each branch and agreeing at the cutoff. A continuous function that is non-decreasing on each of finitely many intervals partitioning its domain is non-decreasing, so $R(\rho; \cdot)$ is non-decreasing, and a pointwise maximum of non-decreasing functions is non-decreasing, so $\bar{R}(\rho; \cdot)$ is too. Dividing by the constant M gives $\Lambda(\rho; z_0) \leq \Lambda(\rho; z_1)$ at every ρ, which is first-order stochastic dominance.

Part (iii). Outside region IV the denominator moves as well. There $\rho^m = \hat{\rho}$ and $M(z) = z\big(1 - c/\hat{\rho}(z)\big)$; substituting $\hat{\rho} = \beta^{1/(\sigma-1)} z^{\sigma/(\sigma-1)}$ from (2.13) and using $1 + \sigma/(1 - \sigma) = 1/(1 - \sigma)$ gives $M(z) = z - c\beta^{1/(1-\sigma)} z^{1/(1-\sigma)}$, with derivative $1 - c\beta^{1/(1-\sigma)} z^{\sigma/(1-\sigma)}/(1 - \sigma)$, positive on the stated range. On regions I and II the numerator $\bar{R}$ may then rise with z faster or slower than M does, and Λ need not be monotone. Part (i) is unaffected because Step 4 never used the behaviour of $\bar{R}$ below $\tilde{\rho}$: the only property of M it used was that $M(z_0) \geq R(\rho_1^m; z_0)$, which is the definition of a maximum. $\square$

The support of F when R is not monotone

Suppose R falls somewhere below ρ^m, and let F be given by (2.22). We verify that this is an equilibrium and that it is the only one.

A firm posting ρ earns $[\alpha_1 + 2\alpha_2(1 - F(\rho))]R(\rho)$. Substituting (2.22) gives $\alpha_1 R(\rho^m)R(\rho)/\bar{R}(\rho)$, which by the definition of $\bar{R}$ is at most $\Pi^* = \alpha_1 R(\rho^m)$, with equality exactly when $R(\rho) = \bar{R}(\rho)$.

Every price in the stated support therefore earns Π^* and every other price earns strictly less, so no firm deviates. Conversely, any equilibrium must satisfy (2.20) on its support with the same Π^*, since the top of the support is ρ^m by the argument above; and $\alpha_1 + 2\alpha_2(1-F)$ is non-increasing, so R is non-decreasing there and F is pinned down by (2.22).

Because R is continuous, so is $\bar{R}$, hence so is F: the distribution has no atom, and on the excluded interval F is merely flat. By Lemma 2 the only interior local maxima of R below ρ^m are $\breve{\rho}$ and ρ^*, since the two fixed-spending branches are strictly increasing, so $\bar{R}$ has a closed form: it equals R except on an interval beginning at whichever of those two is a peak and ending where R climbs back to that level. By Lemma 2 this occurs only in case 3 with $\rho^m = \hat{\rho}$; in cases 1 and 2 the schedule R is increasing on $[\underline{\rho}, \rho^m]$ and $\bar{R} = R$. □

Proof of Lemma 5

Substituting $q(\rho)$ from Lemma 1 into $u'(q)/\rho$ and using $u'(q) = q^{-\sigma}$ gives $X^{-\sigma}\rho^{\sigma-1}$ on $[\underline{\rho}, \breve{\rho})$, where $q = X/\rho$; κ on $[\breve{\rho}, \tilde{\rho})$, where $q = (\kappa\rho)^{-1/\sigma}$ so that $u'(q) = \kappa\rho$; $z^{-\sigma}\rho^{\sigma-1}$ on $[\tilde{\rho}, \hat{\rho})$, where $q = z/\rho$; and β on $[\hat{\rho}, \infty)$. Splitting the integral in (2.26) at the three cutoffs gives (2.26).

For (i), $\psi_C = d$ implies $\kappa = \beta$ and hence $\tilde{\rho} = \hat{\rho}$, so the third integral is over a null set. The second and fourth terms then carry the same coefficient β and may be combined, and what remains is

$$1 + \mu = \alpha_0\beta + X^{-\sigma}\int_{\underline{\rho}}^{\breve{\rho}} \big[\alpha_1 + 2\alpha_2(1-F)\big]\rho^{\sigma-1}dF + \beta\int_{\breve{\rho}}^{\rho^m} \big[\alpha_1 + 2\alpha_2(1-F)\big]dF,$$

in which z and a appear only through X: with $\kappa = \beta$ the cutoff $\breve{\rho} = \beta^{1/(\sigma-1)}X^{\sigma/(\sigma-1)}$ is a function of X alone, F and ρ^m depend on (z, a) only through the demand schedule, and by Lemma 1 that schedule is $q = X/\rho$ below $\breve{\rho}$ and $q = (\beta\rho)^{-1/\sigma}$ above it. This is precisely the Euler equation of Head et al. (2012) with z replaced by X, so its solution $\bar{X}$ is a function of μ alone and $z = \bar{X} - dA$.

For (ii), $\rho^m = \hat{\rho}$ makes the fourth integral one over a null set, since the support of F is $[\underline{\rho}, \rho^m]$.

For (iii), $\breve{\rho} \le \underline{\rho}$ places region I outside the support, so the first integral vanishes. The asset supply enters (2.26) only through X in that term, enters (2.14) only through $\chi = \psi_C a$, which by Lemma 1 is the value of χ on region I alone, and enters (2.27) only through an integral over $\rho < \breve{\rho}$. All three are then free of A, and the last is zero, so $\psi = \beta d$. □

Proof of Proposition 2

(i). By (4.5) the premium is the constant $\beta d\, i$ on $[0, \underline{A}]$, so profit there is $\beta d\, i\, A$ and rises linearly: the monopolist issues at least $\underline{A}$. Past that point ψ_C falls below d and the premium with it, and it falls fast enough that Π turns down — at the parameters of footnote 5 the elasticity $-L'A/L$ is close to four just above the cutoff, so the turn is immediate rather than marginal. The maximum is therefore the kink, and the value there is $\beta d\, i\, \underline{A}(\mu)$.

(ii). Free entry drives the premium to zero, and by Proposition 1(iii) the premium is zero exactly when region I is empty, which is $A \geq \overline{A}(\mu)$. The smallest such supply is $\overline{A}(\mu)$.

(iii). By Proposition 1, $\mathcal{W}$ does not depend on A outside $(\underline{A}, \overline{A})$: below $\underline{A}$ the decentralized market is the benchmark, and above $\overline{A}$ the asset supply has left the equilibrium. Any maximizer outside the band therefore has a counterpart at its edge. □

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